

CROPCAST: THE SMART AGRI-COMMODITY PRICE PREDICTOR

A PROJECT REPORT

Submitted by

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in

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CHENNAI INSTITUTE OF TECHNOLOGY

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Chennai-600 069

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ABSTRACT

Fluctuation in agribusiness commodity prices causes unpredictable levels of stress for consumers, farmers, and policymakers across India, making price stability a persistent national concern. The Department of Consumer Affairs actively monitors 22 essential food commodities on a daily basis—including pulses such as gram, tur, urad, moong, and masur, and vegetables such as onion, potato, and tomato—through nearly 550 reporting centres nationwide. Despite this extensive monitoring mechanism, consistent control over price volatility remains challenging due to inherent seasonality of agricultural commodities, variations in crop yield, disruptions in supply chains, and complex market dynamics involving wages, inflation, exchange rates, and demand–supply imbalances. These factors collectively pose serious risks to food security, economic stability, and rural livelihoods. To address these challenges, this project proposes an AI–ML based predictive forecasting framework for essential pulses and vegetables, with a particular focus on onion, potato, and tomato. Unlike traditional statistical models such as ARIMA that rely primarily on historical price trends and linear assumptions, the proposed approach leverages machine learning and deep learning techniques to capture nonlinear relationships and temporal dependencies. The framework integrates multi-dimensional inputs including seasonality patterns, crop sowing cycles, production estimates, buffer stock releases, market intelligence indicators, and external factors such as weather conditions. Advanced data preprocessing and feature engineering techniques are employed to enhance data quality and model learning capability, followed by hybrid time-series modeling using Long Short-Term Memory (LSTM) networks and Random Forest algorithms to improve prediction accuracy and robustness. The resulting forecasts provide actionable insights that can assist policymakers in designing timely and effective market interventions, optimizing procurement strategies, regulating buffer stock releases, and administering subsidies more efficiently. By translating complex predictive outputs into interpretable and user-centric insights, the system bridges the gap between advanced analytics and real-world agricultural decision-making. Overall, this AI–ML driven initiative strengthens agricultural market intelligence, supports price stabilization efforts, enhances food security, and contributes to a more resilient, data-driven, and sustainable agricultural ecosystem aligned with the goals of Agriculture, Food Technology, and Rural Development in India.

Key Words:

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SYMBOLS & ABBREVIATIONS

Acronym	Expansion
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
ARIMA	Auto-Regressive Integrated Moving Average
LSTM	Long Short-Term Memory
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
TDNN	Time-Delay Neural Network
VAR	Vector Auto Regression
XG Boost	Extreme Gradient Boosting
CNN	Convolutional Neural Network
TCN	Temporal Convolutional Network
GRU	Gated Recurrent Unit
TFT	Temporal Fusion Transformer
ETL	Extract, Transform, Load
FCI	Food Corporation of India
GDP	Gross Domestic Product

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Agriculture has always been the backbone of the Indian economy, contributing significantly to national GDP and rural livelihoods. With nearly 60% of the population dependent on agriculture, the stability of food prices plays a critical role in ensuring food security and sustainable development. Among agricultural commodities, pulses and vegetables such as onion, potato, and tomato are considered essential due to their nutritional value and widespread consumption.

However, these commodities are highly prone to price volatility, driven by seasonal cycles, production variability, storage limitations, transportation inefficiencies, and unpredictable weather conditions. This volatility adversely affects farmers, traders, consumers, and policymakers alike.

The Department of Consumer Affairs currently monitors the daily prices of 22 essential food commodities through a network of 550 reporting centers across India. To stabilize market conditions, the government maintains buffer stocks of key commodities like pulses and onions. Despite these interventions, the unpredictability of prices remains a significant issue due to factors such as:

- Seasonal fluctuations in production
- Crop yield variability caused by weather conditions
- Market demand and supply imbalances
- Transportation and storage inefficiencies
- Policy interventions and global trade influences

With the rise of Artificial Intelligence and Machine Learning, there is an opportunity to develop advanced, data-driven models that can predict prices more accurately, helping farmers plan production, policymakers regulate supply, and consumers access affordable food. Agriculture is not only a source of livelihood for a majority of Indians but also a key sector that directly impacts food security, economic growth, and national stability. India, being one of the largest producers of pulses and vegetables globally, plays a significant role in the global food supply chain.

Despite this advantage, one of the critical challenges faced by the agricultural ecosystem is the instability of commodity prices. Vegetables like onion, potato, and tomato, along with pulses such as gram, tur, urad, moong, and masur, are consumed daily across all socioeconomic sections of the society. Even small fluctuations in their prices can lead to large-scale inflationary impacts, directly affecting household budgets and policymaking. Hence, predicting and stabilizing these prices is of national importance.

1.1.1 Global Context

Globally, agricultural markets are becoming increasingly interconnected and volatile due to climate change, globalization, and shifting demand patterns. Studies from the Food and Agriculture Organization (FAO) indicate that extreme weather events and supply chain disruptions have been major drivers of food price spikes in the last two decades. Price prediction, therefore, is not just a local concern but a global research priority.

1.1.2 Indian Context

In India, price rise of onions or pulses has often led to political and economic repercussions. For example, onion shortages have historically contributed to inflation spikes and have influenced government decisions on exports and imports. This demonstrates how essential commodities are directly linked to policy, economy, and public welfare.

1.2 PROBLEM STATEMENT

The agricultural sector in India forms the backbone of the nation's economy and food security, yet it remains highly vulnerable to the challenge of unpredictable price fluctuations in essential Agri-horticultural commodities such as pulses and vegetables (notably onion and potato).

Price instability in these commodities is not just an economic concern but a social one, as it directly impacts farmers' livelihoods, consumer affordability, and government policy decisions. Even minor variations in weather conditions, pest attacks, transportation bottlenecks, or shifts in market demand can lead to sudden spikes or crashes in commodity prices. This volatility has severe ripple effects: farmers often fail to secure fair returns for their produce, consumers struggle with inflated costs of daily essentials, and policymakers face mounting pressure to stabilize markets and ensure food availability.

Currently, the Department of Consumer Affairs attempts to mitigate these challenges by monitoring daily prices across multiple reporting centers, maintaining buffer stocks, and implementing interventions. However, these measures remain largely reactive in nature and often fall short of preventing crises. Traditional statistical models, such as ARIMA, though historically useful, are inadequate in today's context. These models struggle to capture the non-linear, multi-factor, and seasonal characteristics of agricultural price movements, especially in a country as diverse and complex as India. They also fail to integrate diverse data streams such as weather forecasts, soil health data, transportation costs, and global commodity trends, which play a significant role in determining prices.

The lack of robust and proactive forecasting mechanisms not only perpetuates uncertainty but also undermines long-term planning for all stakeholders. Farmers are unable to make informed sowing and selling decisions, leading to either surplus production and wastage or shortages that drive inflation. Consumers bear the brunt of rising costs of food items, which form a large proportion of household expenditure, especially in lower-income groups. For policymakers, inaccurate or delayed forecasts make it difficult to design effective procurement, subsidy, and buffer stock strategies, thereby affecting national food security and economic stability.

In this context, there is an urgent need to leverage advanced technologies, particularly Artificial Intelligence (AI) and Machine Learning (ML), to build predictive models capable of capturing the complex interplay of factors influencing agricultural prices. Unlike traditional methods, AI-ML based models can process vast amounts of heterogeneous data, identify hidden patterns, and continuously learn from new inputs. By integrating historical price trends, market intelligence, weather and climate variables, production levels, transportation and logistics costs, and even global market indicators, these models can generate more accurate, timely, and actionable forecasts. Such forecasts would empower farmers to plan cultivation cycles effectively, enable consumers to anticipate market conditions, and equip policymakers with data-driven insights to implement proactive interventions rather than reactive measures. Therefore, the core problem addressed in this study is the inadequacy of existing forecasting mechanisms in capturing the highly dynamic and multi-factorial nature of agricultural commodity prices in India.

Developing AI-ML based predictive systems that provide reliable price forecasts is essential for ensuring price stability, protecting farmer incomes, safeguarding consumer interests, and ultimately strengthening the nation's food security framework.

1.2.1 Our Understanding of the Problem

Agricultural commodity prices in India, especially for pulses and vegetables like onion and potato, are highly volatile and unpredictable. This volatility arises from a combination of natural, infrastructural, and policy-related factors. Farmers, consumers, and policymakers are all adversely impacted due to this instability, and existing forecasting methods fail to capture the real complexity of these markets.

Key factors contributing to the problem:

- **Weather Variability:** Unpredictable monsoons, floods, droughts, and heatwaves affect both yield and quality of crops.
- **Production Patterns:** Fluctuations in acreage and shifting cultivation preferences create unpredictable supply levels.
- **Storage Limitations:** Lack of adequate cold storage and warehousing infrastructure leads to wastage during harvest peaks and shortages later.
- **Transport Disruptions:** Poor rural connectivity, strikes, or rising fuel prices often block smooth supply chains.
- **Policy Interventions:** Export bans, MSP announcements, and sudden buffer stock releases cause abrupt market swings.
- **Traditional Forecasting Gaps:** Models like ARIMA can only capture simple trends and seasonality, but fail with non-linear, multi-factor, real-world data.

Impact on stakeholders:

- **Farmers:** Cannot make informed sowing/selling decisions, often forced into distress sales.
- **Consumers:** Suffer from sudden price hikes in essential daily food items.
- **Policymakers:** Struggle to stabilize prices without accurate forecasts, often reacting too late.

1.2.2 Planned Approach

1.2.2.1 Data Collection & Integration

- Gather historical price data of key commodities.
- Include exogenous factors: weather data, production stats, global trade inputs, and buffer stock information.
- Use multiple sources: government databases, open data APIs, and market intelligence reports.

1.2.2.2 Model Development (AI/ML based)

- Apply time-series models enhanced with external features.
- Use ML/DL methods (Random Forests, Gradient Boosting, LSTMs) to capture non-linear, seasonal, and volatile patterns.
- Continuously retrain models with new incoming data for adaptive forecasting.

1.2.2.3 Forecast Generation

- **Short-term forecasts (7–14 days):** For immediate planning in procurement and logistics.
- **Medium-term forecasts (1–3 months):** For policymaking, buffer stock planning, and trade strategies.
- **Long-term forecasts (6–12 months):** For strategic agricultural planning, infrastructure development, crop diversification, and long-term policy formulation.

1.2.2.4 Output Delivery & Visualization

- Interactive dashboards with price trend visualization.
- Real-time alerts (SMS, app notifications) for farmers and stakeholders.
- Scenario-based simulations to test “what-if” conditions for policymakers.

1.2.2.5 User-Centric Framework

- Multilingual, mobile-friendly design for farmer accessibility.
- Community-driven UX to ensure adoption in rural and urban settings alike.

1.3 AGRICULTURAL PRICE VOLATILITY

Price fluctuations in agricultural commodities are influenced by several interconnected factors:

- ❖ **Seasonality:** Many crops are harvested at specific times, creating periods of surplus and shortage.
- ❖ **Weather & Climate:** Unpredictable monsoons, floods, or droughts directly affect yield.
- ❖ **Storage & Supply Chain:** Lack of cold storage and inefficient logistics worsen the mismatch between supply and demand.
- ❖ **Government Interventions:** Imports, exports, subsidies, and buffer stock releases impact short-term price movements.

1.4 OBJECTIVES

The primary aim of this project is to develop an AI-ML based predictive model for forecasting the prices of essential agri-horticultural commodities such as pulses and vegetables (onion, potato, etc.). These objectives are designed to address the limitations of existing systems and provide accurate, reliable, and real-time insights for decision-makers.

1.4.1 General Objective

To design and implement a robust AI-ML framework capable of forecasting short-term and long-term prices of key agricultural commodities with high accuracy.

1.4.2 Specific Objectives

1.4.2.1 Data Collection and Preprocessing

- ❖ To gather historical price data from government sources, wholesale markets, and consumer affairs reports.
- ❖ To preprocess and clean the dataset for handling missing values and outliers.
- ❖ To integrate supplementary data such as weather conditions and production estimates to enrich the dataset.
- ❖ To perform data normalization and temporal alignment to ensure consistency across multiple data sources.

Table 1.1 Descriptive Statistics of Collected Commodity Price Data

Commodity	Data Range	Total Records	Missing Values (%)	Sources
Onion	2010 - 2023	4800	2.3	Govt, Mandis
Potato	2011 - 2023	4650	1.8	Govt, FCI
Gram Dal	2012 - 2023	4500	3.1	Govt Reports
Urad Dal	2011 - 2023	4700	2.7	Mandis, FCI

1.4.2.2 Feature Engineering

- To identify and incorporate key influencing factors such as seasonality, rainfall, buffer stock levels, imports/exports, and demand-supply gaps.
- To transform raw data into meaningful inputs for machine learning algorithms.

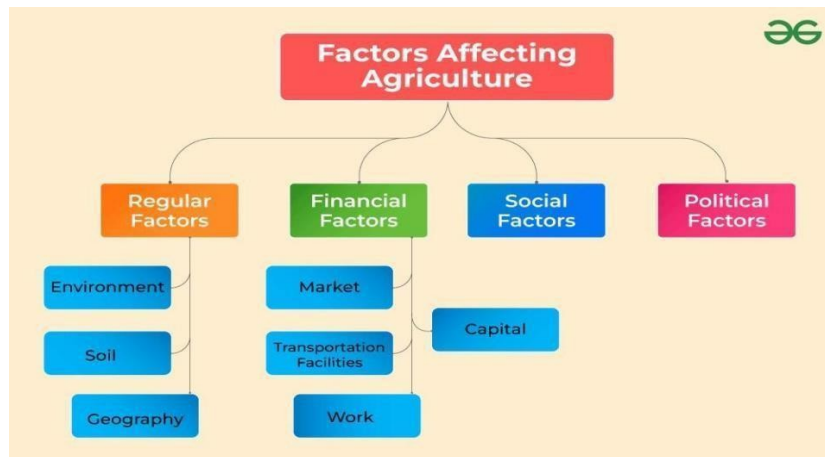


Figure 1.1 Influencing Factors for Price Prediction

1.4.2.3 Model Development and Training

- To experiment with multiple algorithms such as ARIMA, LSTM, Random Forest, and Prophet.
- To train and validate models on historical datasets, ensuring high predictive performance.
- To optimize hyperparameters for better generalization and reduce error rates.

1.4.2.4 Evaluation and Validation

- To measure the accuracy of predictions using evaluation metrics such as RMSE, MAE, and MAPE.

- To benchmark the developed AI-ML models against traditional forecasting techniques like ARIMA.

Table 1.2 Comparative Performance of Forecasting Models on Price Prediction

Model	RMSE	MAE	MAPE (%)	Remarks
ARIMA	15.6	12.1	18.5	Limited for seasonality
Random Forest	11.2	9.4	13.2	Good for nonlinear trends
LSTM	8.9	7.5	10.1	Best for sequential data
Prophet	10.3	8.9	12.4	Strong for trend capture

1.4.2.5 Web Application Overview

The final objective is to provide stakeholders with a user-friendly web application that translates complex forecasting outputs into simple, actionable insights. The application will have a clean and intuitive interface, structured around:

- ❖ **Dashboard Home:** A summary view displaying current prices, recent trends, and forecast highlights.
- ❖ **Interactive Charts & Graphs:** Visual representation of price forecasts (short-term and medium-term) with filters for commodity, region, and time horizon.
- ❖ **Scenario Analysis Module:** “What-if” exploration (e.g., effect of rainfall deficit or buffer stock release on prices).
- ❖ **Stakeholder Views:** Farmers receive a simplified price outlook with short-term (1–2 week) trend predictions to support planning decisions. Consumers are provided with retail price alerts and affordability indicators to assist in informed purchasing.
- ❖ **Notifications & Alerts:** Automated updates via dashboard widgets and optional SMS/Email integration.

This structure ensures that the system is not only analytically robust but also **accessible, scalable, and actionable**, serving the needs of diverse users from farmers to government decision-makers.

1.4.2.6 Deployment and Practical Use

- ❖ To build a user-friendly system/dashboard for real-time visualization of price forecasts.
- ❖ To assist policymakers in buffer stock management and farmers in crop planning.
- ❖ To make forecasting results accessible to stakeholders via web applications.

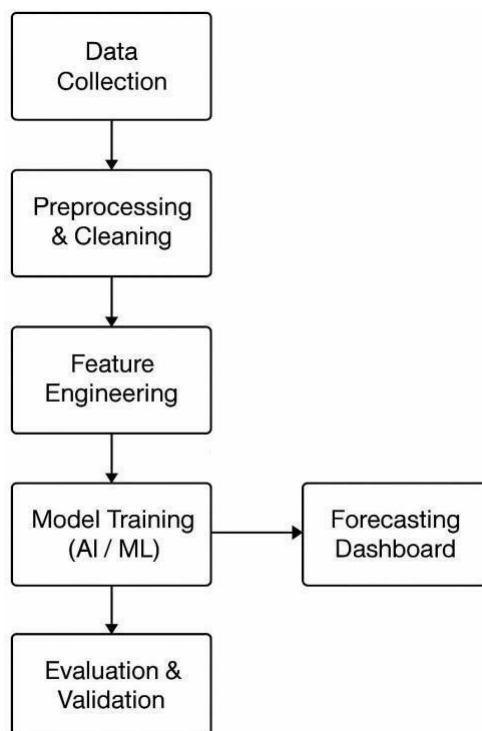


Figure 1.2 Proposed System Objectives

1.4.2.7 Expected Outcomes

- ❖ Reliable price predictions with improved accuracy compared to existing systems.
- ❖ Better decision-making support for government agencies, farmers, and consumers.
- ❖ Reduced price volatility through timely interventions and informed market strategies.

1.5 IMPORTANCE OF PRICE PREDICTION IN AGRICULTURE

Accurate price forecasting of agri-horticultural commodities such as pulses and vegetables is essential because it benefits multiple stakeholders:

Table 1.3 Importance of Price Forecasting for Different Stakeholders

Stakeholder	Importance
Farmers	Enables better crop planning, reduces losses, improves income predictability
Consumers	Ensures availability of affordable essential commodities
Policymakers	Helps design effective interventions like buffer stock release, import/export
Traders	Improves procurement, storage, and distribution strategies
Economy	Strengthens food security and stabilizes inflation

By enabling evidence-based decision-making, price forecasting supports national goals of food security and rural development.

1.6 ROLE OF AI AND ML IN PRICE PREDICTION

The complexity of agricultural price prediction demands methods that can handle large-scale, diverse, and often noisy datasets. Traditional models (e.g., ARIMA, exponential smoothing) rely heavily on historical prices and assume linearity. However, agricultural systems are non-linear, seasonal, and influenced by multi- dimensional factors. AI and ML techniques offer several advantages:

- ❖ **Data Integration:** AI models can incorporate diverse datasets, including weather, soil health, market demand, and policy changes.
- ❖ **Pattern Recognition:** Machine learning algorithms identify hidden patterns that humans or traditional models may overlook.
- ❖ **Adaptability:** Models can update themselves dynamically as new data is fed.
- ❖ **Predictive Accuracy:** Advanced methods such as LSTM neural networks are specifically designed for time-series data like commodity prices.

- ❖ **Decision Support:** AI-based tools can provide not only forecasts but also insights on why a price fluctuation is expected (feature importance).

1.7 SIGNIFICANCE

The importance of this project lies in its ability to provide real-world impact:

- ❖ Farmers gain decision support for crop selection and timing.
- ❖ Policymakers can avoid panic-driven interventions and design evidence-based strategies.
- ❖ Consumers are safeguarded against sudden price surges.
- ❖ The agricultural economy moves towards a data-driven ecosystem.

1.8 SCOPE

The primary goals of this project include:

- ❖ To analyze the historical price trends of pulses and vegetables across different markets.
- ❖ To identify major influencing factors such as seasonality, buffer stock release, crop production, and weather conditions.
- ❖ To develop and implement AI/ML models (Random Forest, LSTM, Ensemble Models) for price prediction.
- ❖ To compare the performance of AI/ML models with traditional approaches like ARIMA.
- ❖ To provide a decision-support system that aids farmers, policymakers, and traders.
- ❖ To forecast the price of pulses and vegetables (onion, potato, tomato, etc.) using AI/ML models with data collected from reporting centers.

1.9 LIMITATIONS OF THE EXISTING SYSTEM

Despite continuous monitoring and forecasting efforts, existing agricultural price prediction systems face several challenges that limit their effectiveness in real-world scenarios. These limitations highlight the need for advanced, data-driven approaches capable of handling complex market dynamics and supporting informed decision-making.

- Existing agricultural price forecasting systems primarily rely on traditional statistical models such as ARIMA, which are limited in capturing nonlinear price behavior and complex market dynamics.
- Most systems depend heavily on historical price data and fail to incorporate multidimensional factors such as weather conditions, production estimates, and market intelligence.
- Conventional approaches struggle to adapt to sudden market disruptions caused by supply-chain issues, policy interventions, or climatic variations.
- Limited scalability and lack of real-time decision support restrict their practical usability for farmers and policymakers.
- Existing systems often provide static forecasts without interactive visualization or user-specific insights, reducing their effectiveness in real-world applications.

1.10 OVERVIEW OF THE PROPOSED SYSTEM

The proposed system is designed to overcome the limitations of existing approaches by leveraging advanced AI and machine learning techniques. It provides a comprehensive, data-driven forecasting framework that integrates multiple influencing factors to deliver accurate and actionable price predictions.

- Utilizes AI and machine learning techniques to accurately forecast agricultural commodity prices.
- Integrates multi-source data including historical prices, weather conditions, production estimates, and market intelligence.
- Employs a hybrid forecasting approach combining LSTM and Random Forest models for improved accuracy and stability.
- Provides interactive visualizations and a web-based dashboard for easy interpretation of results.
- Supports timely decision-making for farmers, consumers, and policymakers through actionable insights.

In summary, the volatility of agricultural commodity prices continue to pose challenges for farmers, consumers, policymakers, and the overall economy. By leveraging historical data, seasonal patterns, market dynamics, and external factors, this project aims to design a robust forecasting framework that not only enhances prediction accuracy but also supports informed decision-making across the agricultural ecosystem.

CHAPTER 2

LITERATURE REVIEW

This chapter provides an overview of significant research carried out in the domain of agricultural commodity price forecasting. Various statistical, machine learning, and deep learning approaches have been explored by researchers worldwide to improve accuracy in predicting market prices. The following review highlights some of the most relevant and original works, detailing their methodology, findings, and limitations, while also positioning how our proposed system builds upon them.

2.1 EXISTING RESEARCH

2.1.1 Forecasting International Maize Prices Using Long Short-Term Memory Networks – Ronit Jaiswal, Girish K. Jha, Kapil Choudhary, Rajeev Ranjan Kumar (2023)

In 2023, Ronit Jaiswal, Girish K. Jha, Kapil Choudhary, and Rajeev Ranjan Kumar applied Long Short-Term Memory (LSTM) networks to forecast international monthly maize prices. Their study highlighted how LSTM models were able to capture both short-term fluctuations and long-term patterns far more effectively than conventional statistical approaches.

The results demonstrated superior accuracy and robustness, making LSTM a promising technique for commodity price forecasting. However, the work was largely restricted to a single commodity and lacked multivariate input factors, limiting its general applicability across diverse agricultural products.

2.1.2 Comparative Analysis of ARIMA, Time-Delay Neural Networks, and LSTM Models for Forecasting Indian Commodity Spot Prices – Manogna R. L., Aswini K. Mishra (2021)

Manogna R. L. and Aswini K. Mishra (2021) carried out a comparative study using ARIMA, Time-Delay Neural Networks (TDNN), and LSTM models for forecasting daily spot prices of Indian commodities such as cottonseed, castorseed, rapeseed, soybean, and guar seed. Their findings revealed that LSTM consistently achieved higher directional accuracy than ARIMA and TDNN, indicating its superiority in capturing price variations.

While this work emphasized the strength of deep learning approaches in Indian agricultural markets, the study was still limited in scope to a few commodities and primarily focused on short-term forecasting.

2.1.3 Machine Learning Approaches for Forecasting Wholesale and Retail Prices of Onion, Potato, and Tomato in India – H. P. Suresha, Mutturaj Uppaladinni, Krishna Kumar Tiwari (2021)

A notable study in 2021 by H. P. Suresha, Mutturaj Uppaladinni, and Krishna Kumar Tiwari focused on forecasting wholesale and retail prices of onion, potato, and tomato in India, while also incorporating variables such as diesel prices and arrival volumes. They compared multiple techniques including Auto ARIMA, RNN, LSTM, VAR, Random Forest Regression, and XGBoost.

The research found that machine learning approaches such as LSTM and XGBoost consistently outperformed traditional statistical models, demonstrating their ability to handle nonlinear patterns in agricultural data. Despite these promising results, the study was limited to selected commodities and did not extend its findings to pulses and other vegetables of national importance.

2.1.4 Deep Learning Models for Forecasting Weekly Onion Prices in Major Indian Markets – A. Kowsalya, G. Sridevi (2020)

In 2020, A. Kowsalya and G. Sridevi applied several deep learning models, including LSTM with attention, CNN, hybrid CNN-LSTM, and Temporal Convolutional Networks (TCN), to forecast weekly onion prices in major Indian markets such as Delhi, Mumbai, and Bangalore. Their study compared these approaches against conventional techniques such as GARCH, TDNN, and Support Vector Regression (SVR).

The results demonstrated that deep learning models, particularly hybrid ones, offered significantly higher accuracy and stability in volatile markets like onions. Nevertheless, the research was limited to a single commodity and did not incorporate broader market or environmental factors that may influence prices.

2.1.5 Application of LSTM Variants for Agricultural Commodity Price Forecasting: A Study on Rice, Wheat, Gram, Banana, and Groundnut – Murugesan, Mishra, Krishnan (2022)

Murugesan, Mishra, and Krishnan (2022) conducted an extensive study applying five different LSTM variants — vanilla LSTM, bi-LSTM, stacked LSTM, CNN- LSTM, and Conv-LSTM — to forecast price data for rice, wheat, gram, banana, and groundnut spanning 2000 to 2020.

The study revealed that all the LSTM-based models achieved accurate short-term forecasts with very low error metrics such as MAE, MAPE, and MSE. This highlighted the effectiveness of deep learning techniques in agricultural price prediction. However, the research primarily focused on univariate forecasting, without integrating additional features like weather conditions, trade policies, or demand-supply shocks.

2.1.6 A Hybrid Approach for Forecasting Agricultural Commodity Prices Using ARIMA, LSTM, and Random Forest – S. Ray, A. Lama, P. Mishra, T. Biswas, S. S. Das, B. Gurung (2023)

More recently, in 2023, S. Ray, A. Lama, P. Mishra, T. Biswas, S. S. Das, and B. Gurung proposed a hybrid forecasting approach combining ARIMA, LSTM, and Random Forest to model volatile agricultural commodity prices. In this framework, Random Forest was used for selecting the most relevant lag features, which were then utilized by ARIMA and LSTM to generate predictions. Their results showed that this hybrid architecture outperformed standalone models, particularly in handling high volatility and irregular price series. The limitation of this study was that it was tested on a narrow dataset, making it less generalizable for a wide variety of agricultural commodities.

From the reviewed works, it is evident that deep learning models, particularly LSTM and its variants, consistently outperform traditional statistical approaches in capturing the non-linear and volatile nature of agricultural commodity prices. However, most existing studies are constrained either by focusing on a limited set of commodities or by excluding important external factors such as weather, policy, or transportation costs. Our project addresses these gaps by proposing a multivariate, hybrid AI-ML framework tailored specifically to pulses and vegetables, aiming to enhance forecasting accuracy and reliability.

2.2 KEY OBSERVATIONS FROM EXISTING RESEARCH

2.2.1 Traditional Statistical Models

- ❖ Early research relied heavily on linear and time-series models like ARIMA, Exponential Smoothing, and regression-based approaches.
- ❖ These models were effective for short-term predictions under stable conditions but struggled with high volatility, seasonality, and multi-factor influences.

2.2.2 Machine Learning Approaches

- ❖ Recent studies increasingly employ supervised ML algorithms such as Random Forests, Gradient Boosting, and Support Vector Machines to predict commodity prices.
- ❖ These approaches can handle non-linear relationships, incorporate multiple features like weather, production data, and trade information, and adapt to changing market conditions.

2.2.3 Deep Learning & Hybrid Models

- ❖ Deep learning methods, including LSTMs and GRUs, have shown strong performance in capturing sequential dependencies and complex seasonal patterns.
- ❖ Hybrid models combining time-series analysis with ML or deep learning are emerging as effective solutions for multi- horizon forecasting.

2.2.4 Incorporation of External Factors

- ❖ Studies highlight the importance of including exogenous variables: rainfall, temperature, soil health, transportation/storage constraints, and government policies.
- ❖ Models incorporating these factors generally outperform those using historical prices alone, particularly for long-term and medium-term forecasts.

2.2.5 Decision Support & User-Centric Applications

- ❖ Some research has focused on delivering predictions via dashboards, mobile apps, and alert systems for farmers and policymakers.
- ❖ Explainable AI models are gaining attention to ensure trust and adoption among

non-technical users.

2.3 GAPS IN EXISTING KNOWLEDGE

Although significant progress has been made in agricultural commodity price forecasting using statistical, machine learning, and deep learning approaches, several critical gaps remain in existing research. Most studies focus on a **limited set of commodities**, often analyzing a single crop or a small group of products, which restricts the generalizability of the proposed models across diverse agricultural markets.

A major limitation observed is the **predominant reliance on univariate or partially multivariate models**, where forecasting is primarily based on historical price data. While some studies incorporate external variables, comprehensive integration of **multi-source factors** such as weather conditions, production estimates, market arrivals, buffer stock interventions, and policy-related variables is still limited. This reduces the ability of existing models to fully capture real-world market dynamics.

Furthermore, many deep learning-based approaches, particularly LSTM and its variants, emphasize prediction accuracy but offer **limited interpretability**, making them less suitable for policy-driven decision support systems. The lack of explainable insights hinders adoption by policymakers and other non-technical stakeholders who require transparency in forecasting outputs.

Another notable gap is the **insufficient focus on hybrid modeling frameworks** that effectively combine temporal learning capabilities of deep learning models with the feature interpretability and robustness of machine learning techniques. While some hybrid models have shown promise, their application is often constrained by narrow datasets and limited validation across varying market conditions.

Finally, relatively few studies address the **deployment and usability aspects** of forecasting systems. Many works stop at model development without providing scalable, user-centric platforms such as dashboards or alert systems that translate predictions into actionable insights. This gap limits the practical applicability of research outcomes in real-world agricultural market management. The proposed work aims to bridge these gaps by developing a **multivariate, hybrid AI–ML forecasting framework** for essential pulses and vegetables, integrating diverse external factors and delivering predictions through an interactive, decision-support-oriented web platform.

2.4 VALUE ADDITION OF THE PROPOSED SYSTEM

The proposed project adds significant value to existing agricultural commodity price forecasting systems by addressing critical limitations identified in prior research and practical implementations. Unlike traditional approaches that rely primarily on historical price trends, this system integrates multi-source data including weather conditions, production estimates, market intelligence, and buffer stock information, enabling a more comprehensive understanding of price dynamics.

The use of a hybrid AI–ML framework combining Long Short-Term Memory (LSTM) networks and Random Forest models enhances both prediction accuracy and robustness. While LSTM effectively captures temporal dependencies and seasonal patterns, Random Forest improves model interpretability and handles nonlinear relationships, resulting in more reliable forecasts under volatile market conditions.

Another key value addition lies in the system’s practical orientation. By deploying the forecasting models through a web-based dashboard with interactive visualizations, alerts, and role-based views, the project translates complex analytical results into actionable insights. This improves accessibility for farmers, consumers, and policymakers, bridging the gap between research and real-world decision-making.

Furthermore, the proposed system supports timely policy interventions such as buffer stock management and price stabilization measures. By enabling short-term and long-term price predictions, the framework assists stakeholders in proactive planning rather than reactive responses. In addition to improved forecasting accuracy and practical deployment, the proposed system is designed with scalability and adaptability in mind. The modular architecture allows seamless integration of additional commodities, regional markets, and real-time data streams as they become available. This flexibility ensures that the framework can evolve with changing agricultural and economic conditions.

Moreover, the inclusion of feature importance analysis and performance monitoring enhances transparency and trust in model predictions, making the system not only technically robust but also suitable for long-term implementation in agricultural market intelligence and policy planning. Overall, the project contributes a scalable, data-driven, and user-centric forecasting solution that enhances market transparency, supports food security, and promotes a sustainable agricultural ecosystem in India.

CHAPTER 3

METHODOLOGY

The present chapter provides a description of the structure, sources of data, steps of data preparation, ways in which to create models (features), and methods of forecasting used to create forecasts for agricultural commodity prices. The proposed solution can be described as a modular, scalable approach to forecasting, that encompasses all the steps required to develop and evaluate forecasting products.

3.1 SYSTEM OVERVIEW

This research focuses on creating accurate price forecasts for the primary agricultural commodities of onion, potato, and tomato using a new predictive methodology that incorporates the following:

- ❖ Historical price data
- ❖ Additional time series price data
- ❖ Real-time data (e.g., weather and crop status)
- ❖ Machine learning and deep learning models
- ❖ Visualization interface

The proposed methodology has the following five high-level components:

- ❖ Data collection
- ❖ Data preparation
- ❖ Feature generation
- ❖ Prediction model development
- ❖ Evaluation and presentation of predictions

The overall system is designed as a modular, end-to-end forecasting framework in which each component operates independently while remaining seamlessly integrated with the others. Data flows sequentially from collection and preparation stages to feature generation and predictive modeling, and finally to evaluation and visualization. This structured architecture ensures scalability, robustness, and ease of interpretation, enabling the system to deliver reliable price forecasts and actionable insights for various stakeholders. The modular design also allows future extensions such as the inclusion of additional commodities and real-time data sources.

3.2 PROBLEM-SOLVING APPROACH

The proposed solution follows a data-driven forecasting approach using artificial intelligence and machine learning techniques. The methodology begins with collecting multi-source agricultural data, followed by data preprocessing and feature engineering to enhance data quality. Machine learning and deep learning models are then trained to forecast commodity prices. A hybrid modeling strategy combining Random Forest and Long Short-Term Memory (LSTM) networks is adopted to improve prediction accuracy and robustness. Finally, the results are evaluated using standard performance metrics and visualized through a web-based dashboard to support decision-making.

3.3 TECHNOLOGIES USED

This project employs a combination of modern data science, machine learning, deep learning, and web development technologies to design and implement an AI-ML based agricultural commodity price forecasting system. The selected technologies support efficient data processing, accurate prediction, and user-friendly visualization of results.

3.3.1 Programming Language

3.3.1.1 Python

- Python is used as the primary programming language due to its simplicity, extensive library support, and strong ecosystem for data science and machine learning.
- It enables rapid development of data preprocessing pipelines, model implementation, and result analysis.
- Python's compatibility with web frameworks allows seamless integration between backend logic and the frontend interface.

3.3.2 Data Processing and Analytics Libraries

3.3.2.1 Numpy and Pandas

- NumPy is used for numerical computations and efficient array operations.
- Pandas is employed for handling structured datasets, time-series manipulation, missing value treatment, and exploratory data analysis.

3.3.2.2 Matplotlib and Seaborn

- These libraries are used to visualize historical trends and forecast results.
- Graphical representations help in validating prediction accuracy.

3.3.3 Machine Learning Frameworks

3.3.3.1 Scikit-learn

- Scikit-learn is used to implement machine learning models such as Random Forest.
- It provides tools for data splitting, feature scaling, model evaluation, and performance metrics.
- Feature importance analysis using Random Forest helps interpret the contribution of different input variables.

3.3.4 Deep Learning Frameworks

3.3.4.1 TensorFlow and Keras

- TensorFlow, along with its high-level API Keras, is used to develop and train Long Short-Term Memory (LSTM) networks.
- LSTM models are effective for time-series forecasting as they capture long-term temporal dependencies and seasonality.
- Keras simplifies model construction, hyperparameter tuning, and training visualization.

3.3.5 Forecasting Models

3.3.5.1 Random Forest Model

- Used to model nonlinear relationships between features and commodity prices.
- Provides robustness against overfitting and allows interpretation of feature importance.

3.3.5.2 Long Short-Term Memory (LSTM) Model

- Applied for sequential price forecasting tasks.
- Capable of learning long-term dependencies and seasonal trends in agricultural price data.

3.3.5.3 Hybrid LSTM–Random Forest Model

- Combines outputs from both LSTM and Random Forest models.
- Improves prediction accuracy and stability under volatile market conditions.

3.3.6 Web Application and Visualization Technologies

3.3.6.1 Streamlit / Flask (Frontend & Backend Integration)

- Used to develop a web-based dashboard for presenting forecasting results.
- Enables interactive selection of commodities and visualization of predicted price trends.

3.3.6.2 HTML, CSS, and JavaScript

- Used to design and enhance the user interface for better usability and responsiveness.
- Supports role-based access and dynamic visualization components.

3.3.7 Data Storage and Management

3.3.7.1 CSV / Local Database

- Used for storing historical datasets and intermediate processed data.
- Enables easy data retrieval and experimentation during model development.

3.3.7.2 Cloud Storage

- Used for scalable storage and deployment.

3.3.8 Development Environment and Tools

3.3.8.1 Jupyter Notebook

- Used for data exploration, preprocessing, model training, and experimentation.
- Provides an interactive environment for iterative development and analysis.

3.3.8.2 Visual Studio Code / PyCharm

- Used for code development, debugging, and project management.

3.3.9 Version Control

3.3.9.1 Git and GitHub

- Used for version control and collaboration.
- Helps track code changes, manage different versions, and maintain project documentation.

3.3.10 Evaluation Metrics and Tools

3.3.10.1 Performance Metrics (MAE, RMSE, MAPE)

- Used to evaluate forecasting accuracy and model reliability.
- Helps compare different models and validate improvements achieved by the hybrid approach.

The integration of Python-based data processing, advanced machine learning frameworks, and web visualization technologies enables the development of a robust and scalable agricultural price forecasting system. These technologies ensure high accuracy, interpretability, and practical usability for real-world decision support.

3.4 SYSTEM ARCHITECTURE

The proposed system architecture is designed as a modular, scalable, and robust framework that seamlessly integrates multiple components, including data acquisition, preprocessing, modeling, and visualization modules, ensuring efficient data flow, high performance, and adaptability to evolving requirements. Furthermore, the architecture supports real-time data processing and continuous model updates, enhancing prediction accuracy and system responsiveness.

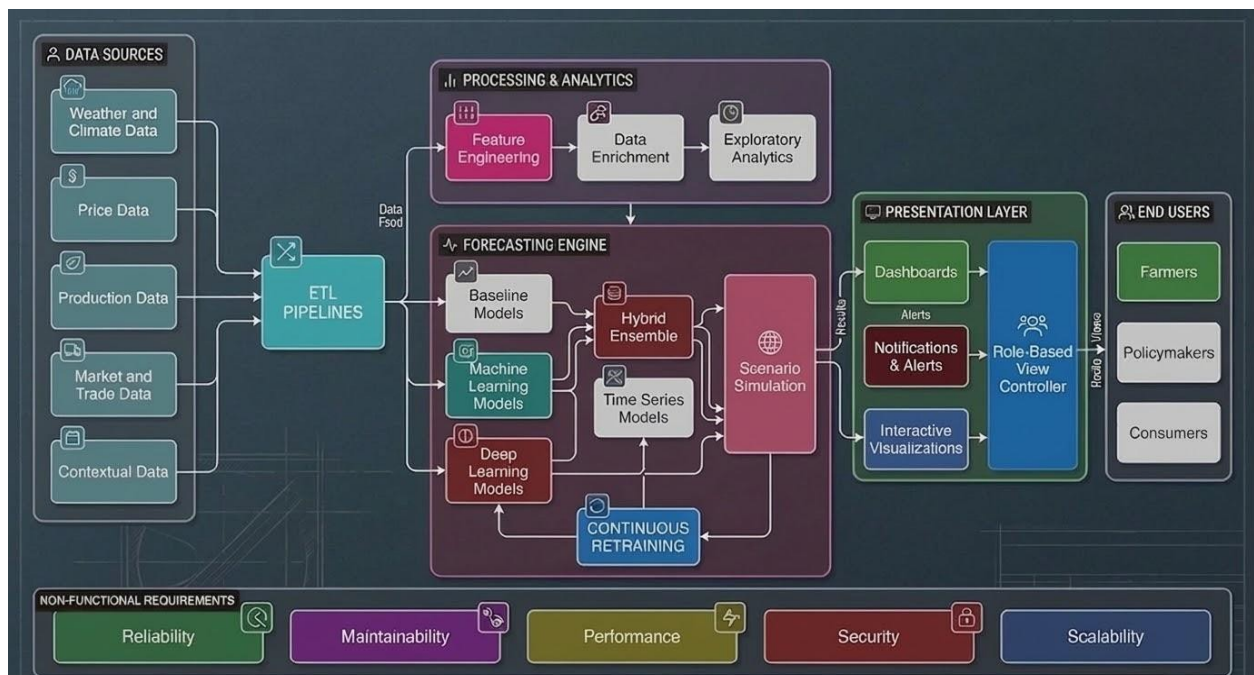


Figure 3.1 End-to-End System Architecture of Hybrid Agricultural Price Forecasting System

The above figure illustrates the end-to-end system architecture of the agricultural commodity price forecasting solution, depicting the systematic flow of data from multiple sources through various analytical and modeling stages to deliver meaningful insights to end users. The architecture integrates diverse data sources, efficient data ingestion mechanisms, advanced preprocessing and analytics modules, and hybrid AI–ML forecasting models to generate accurate price predictions. The processed outputs are delivered through a user-friendly web-based application, enabling interaction with different stakeholders such as farmers, policymakers, and analysts. The overall framework is organized into six major layers, namely **Data Sources, Data Ingestion and Storage, Processing and Analytics, Forecasting Engine, Presentation Layer, and User Layer**, ensuring modularity, scalability, and seamless interaction among system components.

3.4.1 Data Sources Layer

The accuracy and robustness of forecasts depend on the richness and diversity of input data. Our system integrates multiple structured and unstructured sources, ensuring both historical and real-time signals are captured.

- ❖ **Price Data:** Daily/weekly price records from Department of Consumer Affairs monitoring centers across regions.
- ❖ **Agricultural Production Data:** Crop sowing area, yield statistics, mandi arrivals, and procurement records.
- ❖ **Weather & Climate Data:** Rainfall, temperature, drought/flood indices, soil moisture, and long-range IMD forecasts.
- ❖ **Market & Trade Data:** Import/export volumes, buffer stock levels, logistics disruptions, and global commodity indices.
- ❖ **Contextual Data:** Festival periods, consumer demand spikes, news sentiment around crop shortages or bumper harvests.

By combining these inputs, the system captures both **direct and indirect factors** influencing commodity prices.

3.4.2 Data Ingestion and Storage Layer

A **data pipeline** ensures seamless ingestion, cleaning, and organization of raw data into a usable format.

3.4.2.1 ETL Pipeline (Extract, Transform, Load)

- ❖ Automated scripts fetch data from APIs, government reports, and other repositories.

- ❖ Raw data is cleaned (handling missing values, unit conversions, duplicate removal).
- ❖ Transformed into standardized formats for downstream processing.

3.4.2.2 Storage

- ❖ **Data Lake** for storing raw, semi-structured, and unstructured data.
- ❖ **Relational Database (SQL/NoSQL)** for processed, structured datasets ready for model training.
- ❖ **Scalability:** Cloud-based storage ensures data growth is easily managed, and real-time updates are supported.

3.4.3 Processing and Analytics Layer

This layer prepares data for modeling and generates intermediate analytics to support interpretation.

3.4.3.1 Feature Engineering

- ❖ Lag variables (7-day, 14-day, 30-day price trends).
- ❖ Seasonal indicators (harvest months, festivals).
- ❖ Weather-yield lags (rainfall impacting yield with 1–2 month delay).

3.4.3.2 Data Enrichment

Correlating price data with production, trade, and weather signals to enrich the dataset and capture broader market influences.

3.4.3.3 Exploratory Analytics

Visual dashboards to identify correlations, anomalies, and historical price cycles. This ensures the forecasting models are trained on **high-quality, enriched data** that captures both short-term signals and long-term patterns.

3.4.4 Forecasting Engine (AI/ML Layer)

At the heart of the architecture is the forecasting engine, which blends multiple models to provide robust predictions.

- ❖ **Baseline Models:** ARIMA, Prophet — used for benchmarking performance.
- ❖ **Machine Learning Models:** Gradient Boosting (XGBoost, LightGBM) for handling tabular + categorical data.

❖ **Deep Learning Models**

- **LSTM/GRU Networks:** For capturing long-term dependencies in time-series.
- **Temporal Fusion Transformers (TFT):** For multi- horizon forecasting using exogenous variables.

❖ **Hybrid Ensemble:** Combining outputs from multiple models to improve accuracy and reduce bias.

❖ **Scenario Simulation:** Ability to project outcomes under different assumptions (e.g., rainfall deficit, export ban).

The forecasting engine continuously **re-trains** on new data to adapt to changing market dynamics.

3.4.5 Presentation Layer

The forecasting outputs are delivered to end-users via a modern, intuitive web application.

- ❖ **Dashboard:** High-level overview of current prices, historical trends, and forecast ranges.
- ❖ **Interactive Visualizations:** Line charts, bar charts, and heatmaps for commodity and region comparisons.
- ❖ **Scenario Analysis Module:** Allows policymakers to simulate “what- if” scenarios (e.g., effect of releasing X buffer stock).
- ❖ **Role-based Views:** Farmers get a 7–14 day price outlook and consumers receive retail price alerts with budget indicators. Policymakers obtain medium-term insights for buffer stock and trade decisions.
- ❖ **Notifications & Alerts:** Email, SMS, or WhatsApp integration for real- time updates.

The frontend emphasizes **clarity, accessibility, and decision-support**, ensuring insights are both usable and actionable.

3.4.6 System Integration and Workflow

The proposed system architecture follows a structured, layered workflow that ensures efficient data acquisition, processing, analysis, and dissemination of forecasting results. Each layer is designed to perform specific functions while seamlessly interacting with other components, enabling smooth data flow and reliable system performance. This modular design improves scalability, maintainability, and adaptability to changing data sources and analytical requirements. The complete flow of the architecture can be summarized as follows:

3.4.6.1 Data Sources (Price, Weather, Production, Trade, Contextual)

This layer collects data from multiple government portals, market databases, meteorological agencies, and external information systems to ensure comprehensive coverage of factors influencing commodity prices.

3.4.6.2 Data Ingestion (ETL Pipelines → Data Lake + Database)

Raw data is extracted, transformed, and loaded through automated pipelines into centralized storage systems, ensuring consistency, integrity, and availability for downstream processing.

3.4.6.3 Processing & Feature Engineering (Lag Variables, Seasonality, Weather Correlations)

The collected data is cleaned, normalized, and transformed into meaningful features such as lag indicators, rolling statistics, seasonal indices, and weather correlation metrics to enhance model learning capability.

3.4.6.4 Forecasting Engine (ML/DL Ensemble + Scenario Modeling)

This layer applies hybrid machine learning and deep learning models, including Random Forest and LSTM networks, to generate accurate short-term and long-term price forecasts and evaluate alternative market scenarios.

3.4.6.5 Presentation Layer (Web Application + Alerts)

Forecast outputs are visualized through an interactive web-based dashboard and alert mechanisms, enabling users to explore trends, compare scenarios, and receive timely notifications.

3.4.6.6 End Users (Farmers, Policymakers, Consumers)

The final layer delivers customized insights to different stakeholders, supporting informed decision-making, market interventions, and strategic planning.

Overall, this layered architectural design ensures efficient integration of heterogeneous data sources, robust analytical processing, reliable forecasting, and effective communication of results. By combining automation, advanced analytics, and user-centric visualization, the proposed framework provides a comprehensive decision-support platform for agricultural commodity price management.

3.4.7 Non-Functional Considerations

- ❖ **Scalability:** Cloud-native design allows for expansion across new commodities and geographies.
- ❖ **Performance:** Real-time or near-real-time forecasting updates with low latency.
- ❖ **Maintainability:** Modular architecture supports easy updates and integration of new data sources/models.
- ❖ **Reliability:** Automated monitoring of model drift and fallback mechanisms to simpler models in case of anomalies.

In essence, the system architecture ensures that the forecasting framework is **comprehensive, resilient, and impactful**, laying the foundation for long-term food security and market stability.

3.5 DATA COLLECTION

Data is collected from reliable sources such as government agricultural portals, market information systems, and weather datasets. The collected data includes historical commodity prices, production estimates, and weather-related attributes. Multi-source agricultural data is used in the forecasting system to create a more reliable forecasting and more accurate forecast prediction. The datasets that are collected include the following:

- Commodity price historical records
- Production and yield type statistics
- Weather and climate data
- Market intelligence metrics
- Buffer stock and intervention information

Table 3.1 Dataset Description

Datatype	Description	Frequency	Purpose
Price Data	Daily commodity prices	Time-series	Target variable
Production Data	Crop yield & output estimates	Seasonal	Supply estimation
Weather Data	Rainfall, temperature	Daily	Climate impact
Market Data	Demand, arrivals, trends	Periodic	Market behavior
Buffer Stock	Govt. intervention data	Event-based	Policy influence

3.6 DATA PREPROCESSING

Data preprocessing involves handling missing values, removing inconsistencies, and normalizing values to ensure uniformity. Feature engineering techniques such as lag features, rolling averages, and seasonal indicators are applied to capture temporal patterns. Exploratory data analysis is performed to identify trends, seasonality, and price volatility. To ensure model reliability, preprocessing is performed through the following steps:

- ❖ Removal of duplicate records
- ❖ Handling missing values using interpolation or statistical imputation
- ❖ Outlier detection and smoothing
- ❖ Data normalization and scaling
- ❖ Temporal alignment of multi-source datasets

This step ensures consistency and improves convergence during training.

Table 3.2 Preprocessing Techniques

Tasks	Techniques Used	Purpose
Missing values	Mean / interpolation	Completeness
Noise handling	Smoothing	Stability
Normalization	Min–Max scaling	Faster convergence
Date alignment	Time indexing	Temporal consistency
Outlier handling	IQR / thresholding	Robust learning

3.7 FEATURE ENGINEERING

Feature engineering plays a critical role in capturing patterns hidden in agricultural time-series data. The features extracted are:

- ❖ Lagged price features ($t-1$, $t-7$, $t-30$)
- ❖ Moving averages and rolling statistics
- ❖ Seasonal indicators (month, week)

- ❖ Trend components
- ❖ Weather-based attributes
- ❖ Production and buffer-stock indicators

These features help models learn temporal dependencies and external influences.

Table 3.3 Feature Categories

Feature Category	Examples
Lag Features	Price($t-1$), Price($t-7$), Price($t-30$)
Seasonal Features	Month, Week, Seasonal index
Statistical Features	Moving average, variance
Weather Features	Rainfall, temperature
Market Indicators	Arrivals, buffer stock levels

3.8 FORECASTING MODELS AND EXPERIMENTAL SETUP

A combination of Machine Learning & Deep Learning techniques is used to develop a reliable forecasting tool. The experimental setup involves training and testing the forecasting models using historical agricultural data. The dataset is divided into training and testing subsets to validate model performance. For that reason, we have identified 3 Models that will be used in the development of this tool as follows:

3.8.1 Random Forest (RF) Model

The Random Forest Model is the baseline model for Machine Learning because of its robust capabilities to handle non-linear data relationships and to produce feature importance scores for interpretability.

3.8.2 Long Short-Term Memory (LSTM) Model

Long Short-Term Memory Networks (LSTM) are designed to capture long term temporal features and dependencies of prices. LSTM will help model the seasonality and sequential dependencies found in most agricultural related data.

3.8.3 Hybrid Models (LSTM+RF)

The predictions from Random Forest and LSTM will be combined into Hybrid Ensemble Forecasting Models to capitalize on the following:

- ❖ Temporal Learning from LSTM
- ❖ Feature based Learning from Random Forest

The final output of Hybrid Models will produce a more stable forecast accuracy. Model performance is assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE). The experiments are conducted in an offline environment using Python-based frameworks.

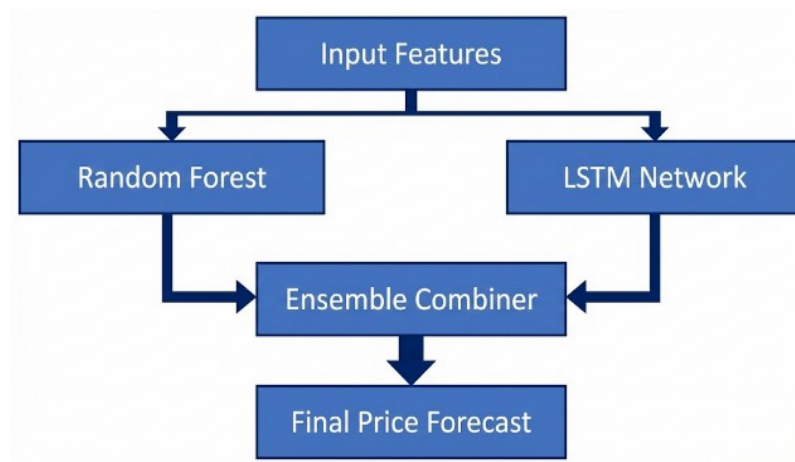


Figure 3.2 Hybrid Forecasting Architecture

3.9 MODEL TRAINING AND EVALUATION

The developed forecasting models were trained and evaluated using historical agricultural price data under a structured experimental setup. The training and evaluation process involves a series of steps.

3.9.1 Data Splitting

The collected dataset was divided into training and testing subsets to ensure an unbiased evaluation of the forecasting models. The training dataset was used to learn underlying patterns in agricultural price data, while the testing dataset was reserved for validating predictive performance on unseen data. This approach helps assess the generalization capability of the models under varying market conditions.

3.9.2 Model Training

Machine learning and deep learning models were trained using historical price data and engineered features. Random Forest models were trained to capture nonlinear relationships among input variables, while LSTM networks were trained to learn temporal dependencies and seasonal patterns. A hybrid model was developed by combining the outputs of both models to improve robustness and accuracy and thus contributing to effective planning and informed decision-making.

3.9.3 Model Evaluation

The trained models were evaluated using standard regression metrics commonly applied in time-series forecasting. **Mean Absolute Error (MAE)** was used to measure average prediction error, **Root Mean Square Error (RMSE)** was employed to penalize larger deviations and assess model stability, and **Mean Absolute Percentage Error (MAPE)** provided a percentage-based measure of prediction accuracy. The use of multiple metrics enabled a comprehensive comparison of model performance.

3.10 OUTPUT GENERATION AND VISUALIZATION

The forecasting outputs were converted into meaningful insights through an interactive and user-friendly visualization layer. The output generation process includes the following components:

3.10.1 Web-Based Dashboard

- ❖ Displays predicted prices along with historical trends.
- ❖ Allows users to select commodities, time horizons, and scenarios.

3.10.2 Interactive Visualizations

- ❖ Time-series plots for actual vs. predicted prices.
- ❖ Comparative views across different forecasting models.
- ❖ Trend and volatility indicators for better interpretation.

3.10.3 Decision-Support Features

- ❖ Scenario-based insights to evaluate potential market outcomes.
- ❖ Role-based access ensures relevant information for different stakeholders.

3.10.4 Usability and Interpretability

- ❖ Simplifies complex model outputs into visual formats.
- ❖ Enhances usability for policymakers, analysts, and decision-makers.

By translating complex analytical results into intuitive visual formats, the system supports informed decision-making and effective policy intervention.

3.11 EXISTING SYSTEM AND PROPOSED WORK

3.11.1 Existing System

Existing agricultural commodity price forecasting systems primarily rely on traditional statistical models such as ARIMA and related time-series techniques. These models are mainly dependent on historical price data and assume linear relationships within the dataset. While such approaches are effective under stable market conditions, they struggle to handle complex, nonlinear interactions and sudden market shocks caused by weather variability, supply-chain disruptions, or policy changes.

Moreover, traditional systems are limited in scope, often providing only short-term forecasts based on past trends and basic seasonality. The static nature of these models requires frequent manual recalibration, reducing their adaptability to rapidly changing market dynamics. As a result, existing systems offer moderate forecasting accuracy and limited decision-support capability for stakeholders.

3.11.2 Proposed Work

The proposed work introduces an AI–ML based forecasting framework designed to overcome the limitations of traditional approaches. By integrating diverse data sources such as historical prices, weather conditions, production estimates, logistics factors, and policy-related information, the system captures a comprehensive view of agricultural market dynamics. Machine learning and deep learning models, including Random Forest and LSTM, are employed to model nonlinear, multi-factor relationships and temporal dependencies.

Unlike static traditional models, the proposed framework supports multi-level forecasting, enabling short-term and medium-term price predictions to meet the needs of different stakeholders. The adaptive nature of the AI–ML models allows continuous learning and improved accuracy under volatile conditions.

Table 3.4 Existing System vs Proposed Work

Factor	Existing System (Traditional Models like ARIMA)	Proposed Work (AI-ML Based Models)
Data Sources	Limited to historical price data, mainly time-series trends	Integrates diverse datasets: historical prices, weather, production, logistics, global trade, and policy interventions
Modelling Capability	Captures only linear patterns, simple trends, and basic seasonality	Handles non-linear, multi-factor, and dynamic relationships using ML/DL models (RF, GBM, LSTM)
Forecast Accuracy	Moderate; fails under sudden shocks, high volatility, or multi-factor influences	Higher accuracy through feature-rich models, continuous learning, and adaptive retraining
Scope of Forecasting	Mostly short-term, limited to basic seasonal patterns	Multi-level: short-term (7–14 days), medium-term (1–3 months), adaptable to diverse needs
Adaptability	Static models; need manual updates and recalibration	Dynamic, self-learning models that update automatically with new data

In summary, the comparative analysis highlights the clear advantages of the proposed AI–ML based forecasting framework over traditional statistical models. By integrating multi-source data and advanced learning techniques, the system overcomes the limitations of linear and static forecasting approaches, resulting in more accurate, adaptive, and scalable price predictions. The hybrid RF–LSTM model effectively captures nonlinear patterns, seasonal trends, and market volatility.

Furthermore, the enhanced predictive capability supports proactive planning and informed decision-making for farmers, consumers, and policymakers. By enabling better anticipation of market fluctuations and timely interventions, the proposed framework contributes to improved market stability, greater transparency, and strengthened food security. It also reduces uncertainty in agricultural markets and helps stakeholders optimize resource allocation. Overall, the system fosters a more resilient, transparent, and data-driven agricultural ecosystem that supports sustainable market growth and informed decision-making.

CHAPTER 4

IMPLEMENTATION AND DEVELOPMENT PROCESS

This chapter describes the implementation of the proposed agricultural commodity price forecasting system in detail. It presents the architectural design, functional workflow, and integration of system components required to transform raw agricultural data into accurate price forecasts and actionable insights. The chapter also explains how individual modules were developed, tested, and integrated to form a scalable, secure, and efficient end-to-end forecasting platform. Supporting system diagrams are referenced to illustrate the data flow and interaction between modules.

4.1 OVERVIEW OF SYSTEM IMPLEMENTATION

The development of the agricultural commodity price forecasting system was carried out using a structured and modular implementation strategy. The system was designed to support continuous data acquisition, automated processing, predictive modeling, and user-oriented visualization. Emphasis was placed on scalability, adaptability, and robustness to accommodate the dynamic nature of agricultural markets and environmental conditions.

Initially, historical datasets were collected from authenticated government portals, agricultural market repositories, and meteorological data providers. These datasets included commodity price records, production statistics, weather variables, and market indicators spanning multiple years. Since the raw data originated from heterogeneous sources, it contained inconsistencies such as missing entries, varying temporal resolutions, and noise. Comprehensive preprocessing techniques were applied to ensure data reliability and consistency across sources.

The preprocessing stage involved data cleaning, handling missing values through statistical and interpolation techniques, normalization of numerical attributes, and alignment of time-series data. Feature engineering was then performed to derive predictive variables such as lagged price values, rolling averages, seasonal indices, and weather-influenced indicators. These engineered features enabled the forecasting models to capture temporal trends, seasonal patterns, and external influences on commodity prices. Subsequently, machine learning and deep learning models were developed and trained using the processed dataset.

A hybrid forecasting approach combining Random Forest and Long Short-Term Memory (LSTM) networks was implemented to leverage both non-linear regression capabilities and sequential pattern learning. Model performance was evaluated using standard error metrics to ensure prediction accuracy and generalization. Finally, a web-based application was developed to present forecasting results through interactive dashboards, thereby enabling informed decision-making for different user groups. Each system module was independently implemented and validated before being integrated into a unified forecasting pipeline.

4.2 SYSTEM MODULES AND IMPLEMENTATION DETAILS

The forecasting system is composed of multiple interconnected modules, each designed to perform a specific function within the overall workflow. These modules collectively facilitate seamless data flow from acquisition to prediction and visualization.

4.2.1 Data Sources Module



The Data Sources Module forms the foundational layer of the system by aggregating diverse datasets that influence agricultural commodity prices. Accurate forecasting requires consideration of multiple external and internal factors, and therefore, data was collected from various domains to ensure comprehensive coverage. Weather and climate data such as rainfall, temperature, humidity, etc, were incorporated due to their direct impact on crop yield and supply.

Figure 4.1 Data Sources Module

Historical price data, market arrivals, trade volumes, and production statistics were also included to capture demand–supply dynamics. In addition to quantitative data, contextual information such as government policy interventions, minimum support prices, and market regulations was considered to reflect real-world influences on agricultural pricing. Data acquisition was performed through a combination of API calls, publicly available datasets, and structured data files. The inclusion of heterogeneous data sources enhances the robustness of the forecasting models by allowing them to learn complex interactions among climatic, economic, and production-related variables.

4.2.2 Data Ingestion and Storage Module

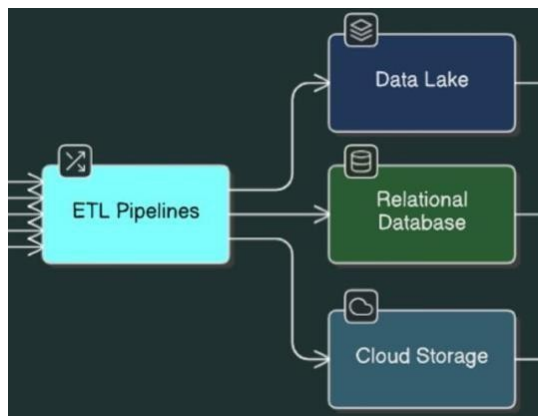


Figure 4.2 Data Ingestion and Storage Module

The Data Ingestion and Storage Module is responsible for collecting and storing raw data in formats suitable for downstream analytics and modeling. This module implements ETL pipelines that automate the extraction of data from external sources, transform it into standardized representations, and load it into persistent storage systems.

Transformation processes include data validation, schema enforcement, type conversion, normalization, and temporal synchronization. The system employs a multi-layered storage architecture to balance scalability, performance, and flexibility. Structured datasets are stored in relational databases to support efficient querying and aggregation, while large historical and semi-structured datasets are maintained in cloud storage and data lake environments. This architecture ensures efficient access to both real-time and historical data, supports large-scale analytics, and enables future system expansion without architectural redesign.

4.2.3 Processing and Analytics Module

The Processing and Analytics Module converts stored raw data into high-quality analytical datasets suitable for forecasting. This module plays a critical role in enhancing data interpretability and predictive relevance.



Figure 4.3 Processing and Analytics Module

Data enrichment is performed by merging datasets from multiple domains, thereby improving contextual depth and reducing informational gaps. Exploratory data analysis techniques are applied to identify trends, correlations, and anomalies within the data. These analyses assist in understanding underlying price dynamics and validating assumptions before model training. The outputs of this module form the final input dataset for the forecasting engine. Feature engineering techniques are applied to derive lag-based features, moving averages, volatility indicators, and seasonal markers that capture recurring patterns in commodity prices. Weather-driven features are also computed to quantify the impact of climatic variability on agricultural output and market behavior.

4.2.4 Forecasting Engine Module

The Forecasting Engine represents the core analytical component of the system. It integrates multiple predictive methodologies to generate accurate and reliable agricultural commodity price forecasts. Baseline statistical models are initially employed to establish reference performance using historical averages and trend analysis. Machine learning models, particularly Random Forest regressors, are then used to capture complex non-linear relationships among features and improve robustness against noise and outliers.

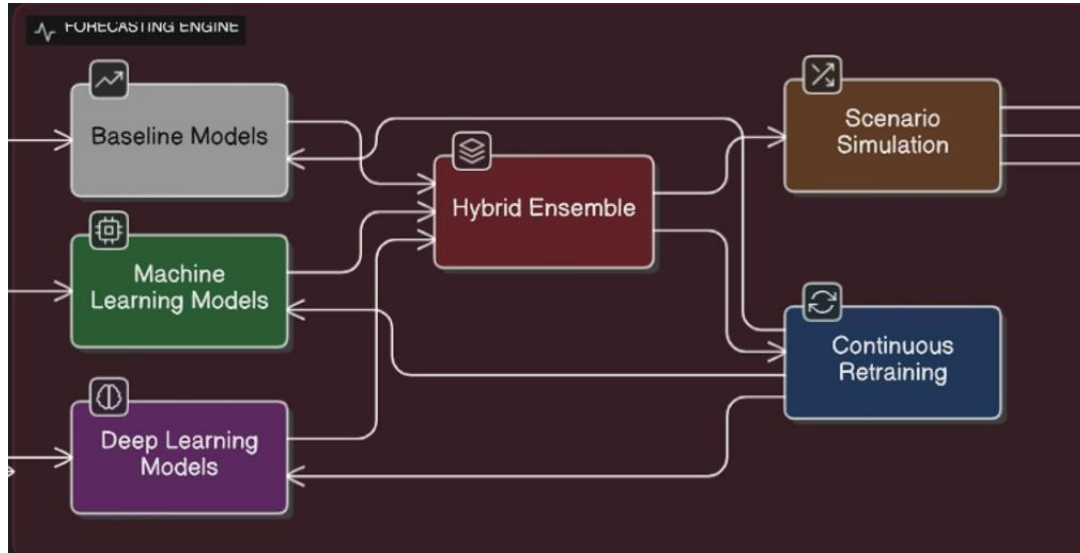


Figure 4.4 Forecasting Module

Deep learning models based on LSTM networks are incorporated to model sequential dependencies and long-term temporal patterns in price data. A hybrid ensemble framework combines the outputs of baseline, machine learning, and deep learning models to enhance forecast accuracy and reduce individual model bias.

The engine supports continuous retraining mechanisms, allowing models to adapt to newly available data and evolving market conditions. Additionally, scenario simulation capabilities enable users to assess the impact of hypothetical changes in weather pattern or policy decisions on future prices.

4.2.5 Web Application Module

The Web Application Module serves as the interface between the forecasting system and its users. It transforms complex analytical outputs into intuitive visual representations that facilitate interpretation and decision-making. The dashboard provides a centralized view of commodity price forecasts, historical trends, and key performance indicators. Interactive charts and visualizations allow users to explore data at different temporal and spatial resolutions.

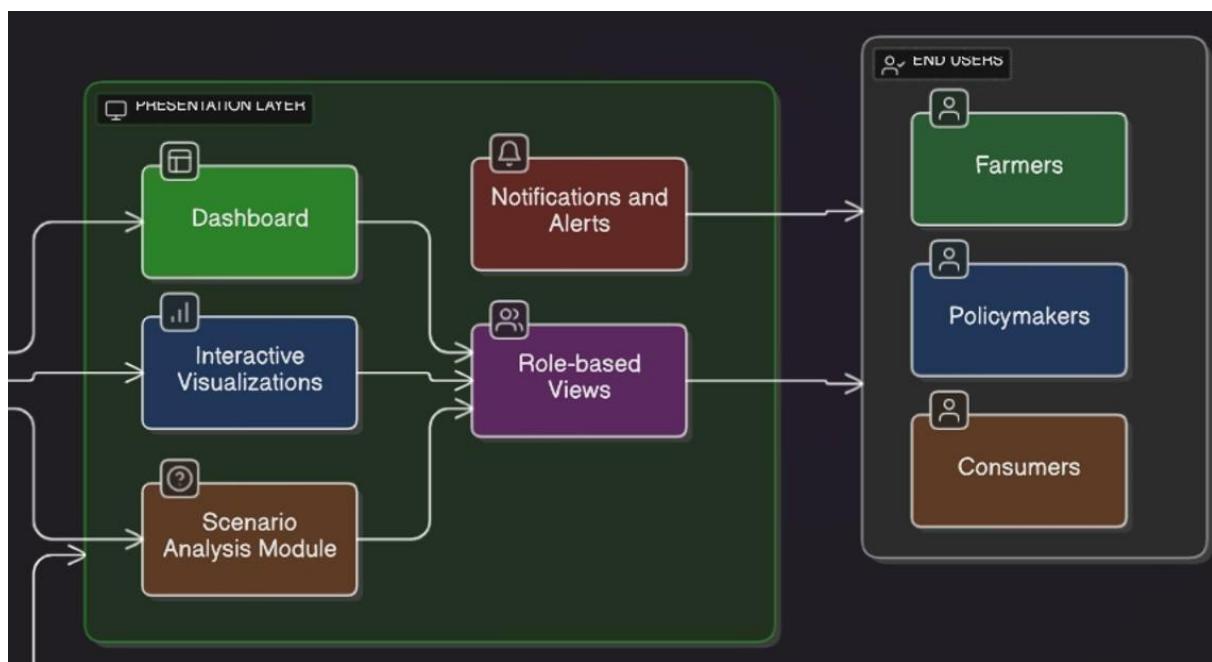


Figure 4.5 Web Application Module and User Modules

The application also incorporates an alert and notification mechanism to inform users of significant price fluctuations, forecast deviations, or risk conditions. Role-based access ensures that users are presented with information relevant to their specific needs. Scenario analysis tools within the interface allow users to interactively evaluate alternative conditions and observe corresponding forecast outcomes. The web application is designed to be responsive, scalable, and accessible across multiple devices.

The dashboard emphasizes clarity and interpretability through consistent visuals and contextual tooltips, translating complex forecasts into actionable insights and enhancing practical usability.

4.2.6 Non-Functional Requirements Module



The non-functional aspects of the system are critical to its practical deployment and long-term usability. Scalability is achieved through modular system design and cloud-based resource allocation, enabling the platform to handle increasing data volumes and user traffic without performance degradation. Security measures such as encryption, authentication, and role-based access control are implemented to protect sensitive data and ensure compliance with data privacy regulations. Performance optimization techniques, including efficient query execution and asynchronous processing, ensure low latency, fast dashboard loading, and timely forecast generation even under high system load.

Figure 4.6 Non-Functional Module

Figures 4.1 through 4.6 illustrate the overall system architecture, demonstrating the end-to-end data flow from acquisition and processing to forecasting and user interaction.

4.3 VISUAL REPRESENTATION OF THE SYSTEM

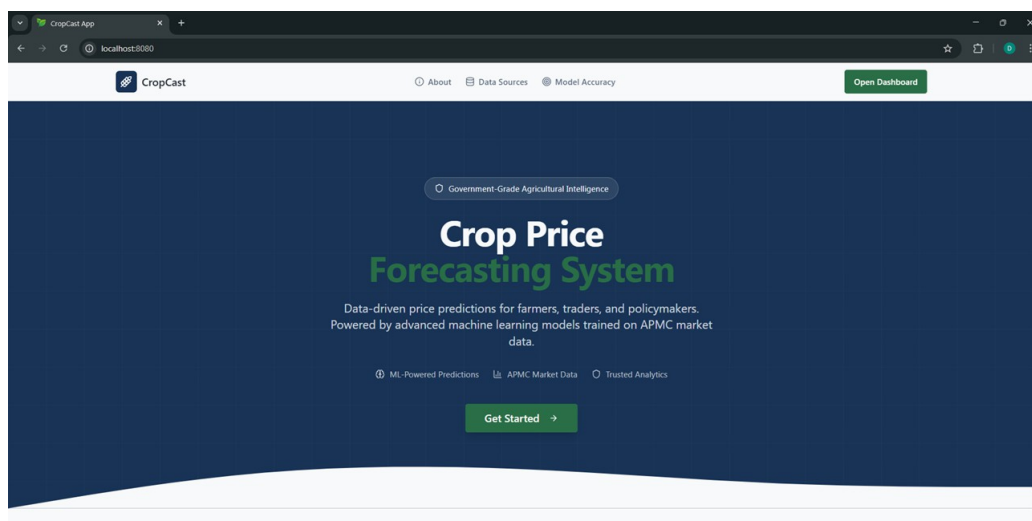


Fig 4.7 Landing Page Interface

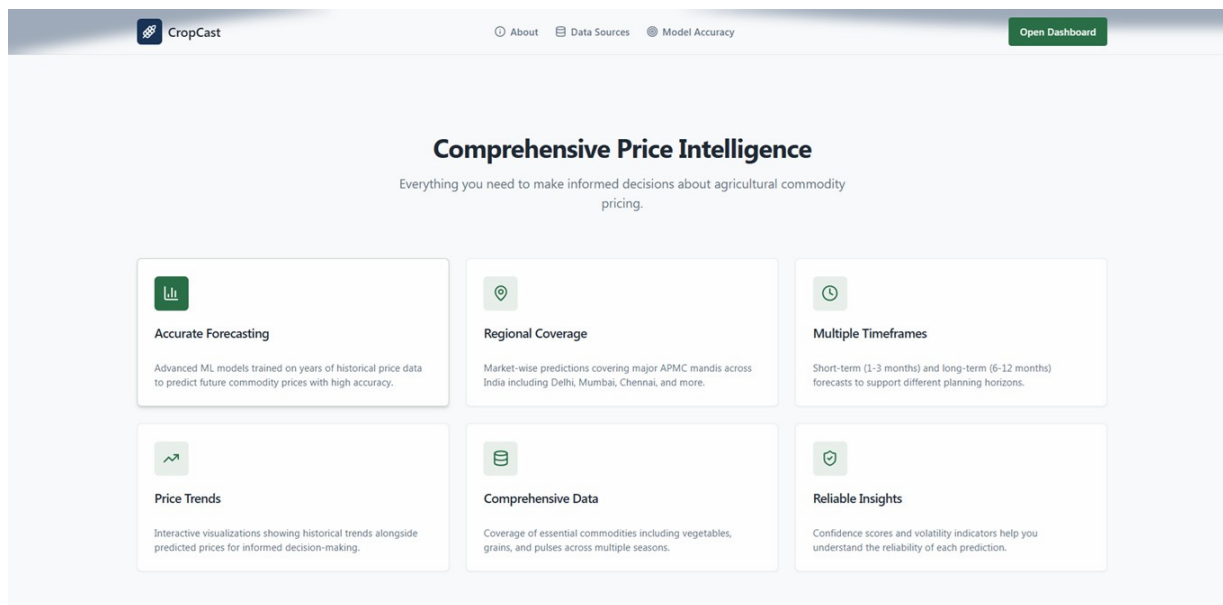


Fig 4.8 Price Intelligence Overview

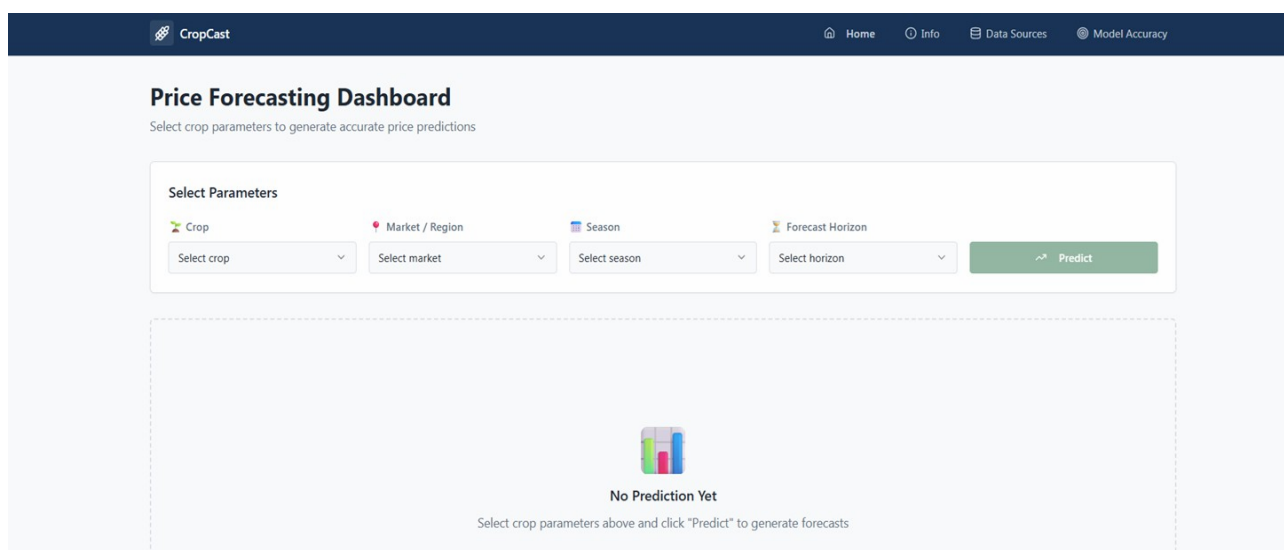


Fig 4.9 Price Forecasting Dashboard Interface

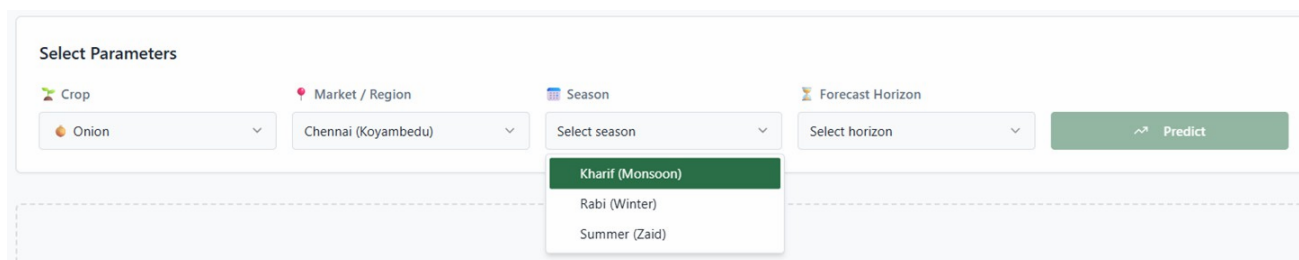


Fig 4.10 Crop, Market and Seasonal Selection Module

Select Parameters

Crop

Onion

Market / Region

Chennai (Koyambedu)

Season

Rabi (Winter)

Forecast Horizon

Select horizon

Short-term (1-3 months)

Long-term (6-12 months)

Predict

Fig 4.11 Forecast Horizon Selection Module

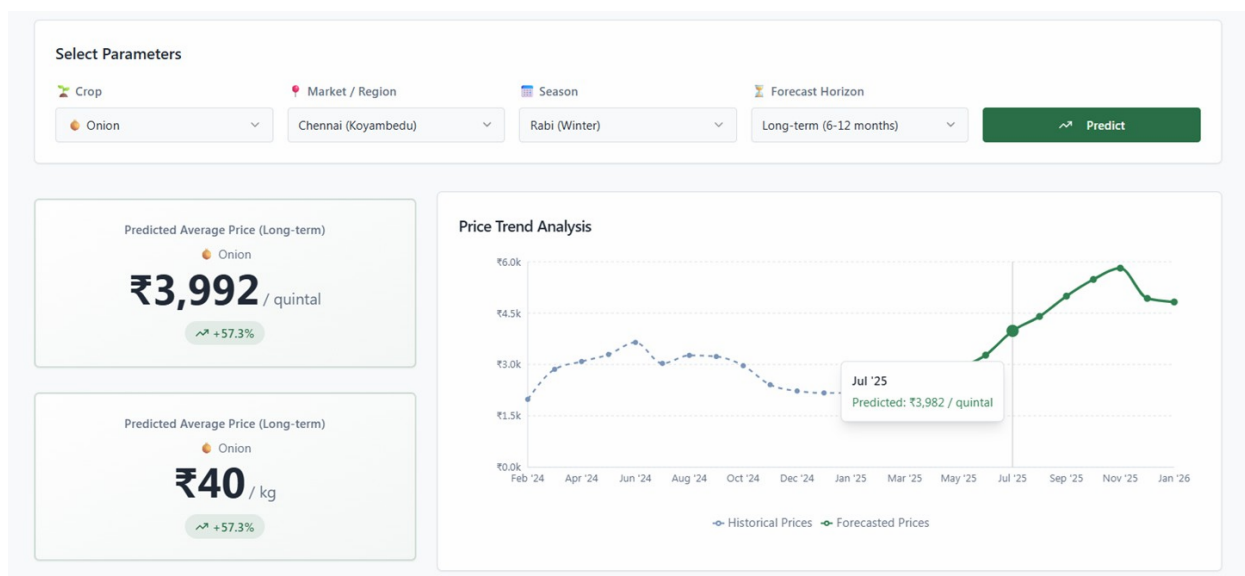


Fig 4.12 Predicted Average Price Summary (Long-Term Forecast)

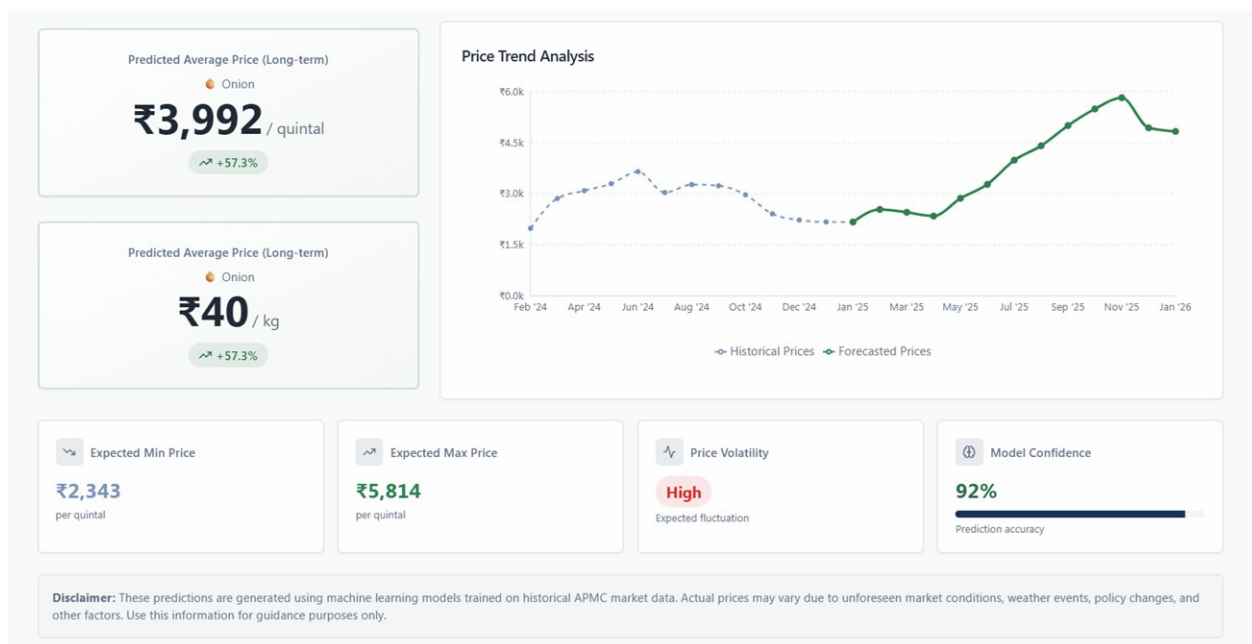


Fig 4.13 Price Trend Analysis

Top Gainers(Current trends)		
Item Name	Price (per Qtl.)	Change
Niger	₹6657.0	6.02% ▲
Gram	₹3771.6	3.38% ▲
Ragi	₹3223.5	3.37% ▲
Copra	₹10934.4	2.39% ▲
Arhar	₹3728.0	1.39% ▲

Top Losers(Current trends)		
Item Name	Price (per Qtl.)	Change
Jute	₹2793.9	-11.42% ▼
Sunflower	₹3722.2	-3.64% ▼
Bajra	₹1502.82	-2.89% ▼
Moong	₹3955.0	-2.46% ▼
Soyabean	₹3348.4	-2.44% ▼

Fig 4.14 Overall Price Trend Analysis

Explore by commodity

 Paddy	 Wheat	 Barley	 Soya Bean
 Cotton	 Coconut	 Ground Nut Seeds	 Mustard Seed
 Gingelly Seed(Sesamum)	 Gram	 Sugarcane	 Arhar
 Ragi	 Maize	 Moong	 Masoor
 Urad	 Raw Jute	 Niger Seed	 Kardi Seed
 Sunflower	 Jowar	 Bajra	

Fig 4.15 Listing of Crops for Price Trends

wheat



Current Price	₹ 1556.55 / ql
Prime Location	U.P., Punjab, Haryana, Rajasthan, M.P., bihar
Crop Type	rabi
Export	Sri Lanka, United Arab Emirates, Taiwan

Brief Forecast

Min. crop price time Oct 26 ₹1476.9

Max. crop price time Dec 26 ₹1672.65

Fig 4.16 Wheat Price Overview

gram



Current Price	₹ 3228.4 / ql
Prime Location	Madhya Pradesh, Maharashtra, Rajasthan, Uttar Pradesh, Andhra Pradesh & Karnataka
Crop Type	rabi
Export	Vietnam, Spain, Myanmar

Brief Forecast

Min. crop price time Oct 26 ₹3063.2

Max. crop price time Dec 26 ₹3469.2

Fig 4.17 Gram Dal Price Overview

bajra



Current Price	₹ 1354.775 / ql
Prime Location	Rajasthan, Maharashtra, Haryana, Uttar Pradesh and Gujarat
Crop Type	kharif
Export	Oman, Saudi Arabia, Israel, Japan

Brief Forecast

Min. crop price time Oct 26 ₹1285.45

Max. crop price time Dec 26 ₹1455.82

Fig 4.18 Bajra Price Overview

soyabean



Current Price	₹ 2536.6 / ql
Prime Location	Madhya Pradesh, Maharashtra, Rajasthan, Madhya Pradesh and Maharashtra
Crop Type	kharif
Export	Spain, Thailand, Singapore

Brief Forecast

Min. crop price time Oct 26 ₹2406.8

Max. crop price time Dec 26 ₹2725.8

Fig 4.19 Soyabean Price Overview

Forecast Trends

Month	Price (per Qtl.)	Change
Mar 26	₹1436.06	-1.03% ▼
Apr 26	₹1452.25	0.09% ▲
May 26	₹1443.53	-0.52% ▼
Jun 26	₹1409.91	-2.83% ▼
Jul 26	₹1385.0	-4.55% ▼
Aug 26	₹1380.01	-4.89% ▼
Sep 26	₹1380.01	-4.89% ▼
Oct 26	₹1362.58	-6.09% ▼
Nov 26	₹1419.87	-2.15% ▼
Dec 26	₹1543.17	6.35% ▲
Jan 27	₹1431.08	-1.37% ▼
Feb 27	₹1451.01	0.0% ▲

Fig 4.20 Month-wise Forecast Trends

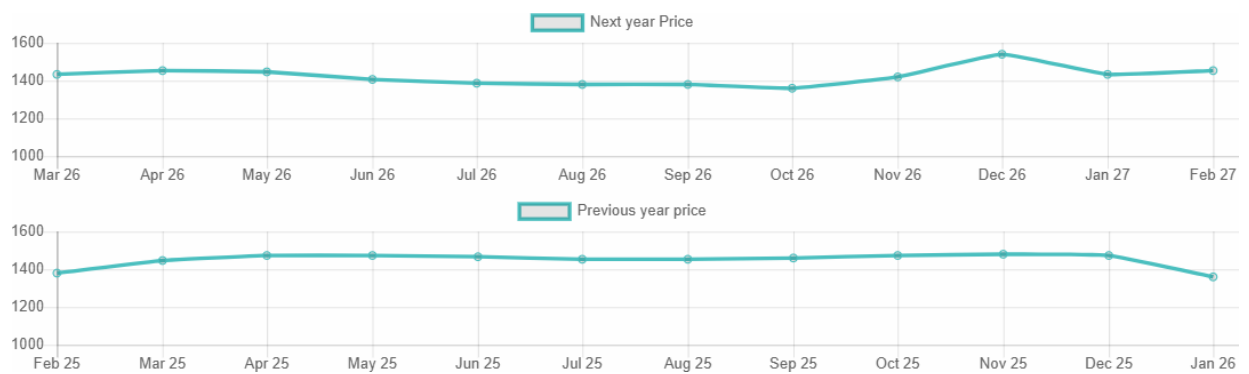


Fig 4.21 Year-wise Forecast Trend Graph

4.4 CHALLENGES AND MITIGATION STRATEGIES

During the development and implementation of the proposed agricultural commodity price forecasting system, several technical and practical challenges were encountered.

Table 4.1 Implementation Challenges and Solutions

Challenge Identified	Impact	Solution Implemented
Data inconsistency and missing values	Reduced data reliability	Applied preprocessing, interpolation, and normalization
Model overfitting and instability	Poor generalization	Used hybrid LSTM–RF approach and validation
Integration of multiple modules	Workflow disruption	Adopted modular system architecture
Dashboard performance issues	Slow user interaction	Optimized data handling and visualization

One of the primary challenges was **data quality and consistency**, as agricultural datasets obtained from multiple sources often contained missing values, inconsistencies, and noise. This issue was addressed through systematic data preprocessing techniques such as interpolation, normalization, and outlier detection, ensuring reliable input for model training. Another significant challenge involved **model overfitting and performance stability**, particularly in deep learning models like LSTM, which are sensitive to parameter selection and data variability. To mitigate this, careful model tuning and validation were carried out, and a hybrid ensemble approach combining Random Forest and LSTM was adopted to improve robustness and generalization.

Integration of multiple system modules also posed challenges, as data needed to flow seamlessly from collection and preprocessing stages to forecasting and visualization components. This was resolved by adopting a modular system architecture, enabling independent development and easier integration of individual modules. Additionally, **performance and responsiveness** of the web-based dashboard were considered, especially when handling multiple user interactions and data visualizations. Efficient data handling techniques and optimized visualization components were implemented to ensure smooth operation. Overall, addressing these challenges contributed to improving system reliability, scalability, and usability, resulting in a robust and effective forecasting framework.

CHAPTER 5

TESTING AND VALIDATION

CHAPTER 6

RESULTS AND DISCUSSIONS

This chapter presents the experimental outcomes of the proposed AI–ML based agricultural commodity price forecasting framework. The performance of the system is evaluated based on prediction accuracy, model stability, and the ability of the hybrid approach to capture complex relationships between commodity prices and influencing factors.

5.1 EXPERIMENTAL OBSERVATIONS AND ANALYSIS

5.1.1 Experimental Setup

To evaluate the forecasting capability of the proposed framework, datasets containing historical crop prices and weather-related information were divided into training and testing subsets. Two forecasting models were developed and evaluated:

- **Random Forest (RF) Model:** An ensemble machine learning technique that constructs multiple decision trees and averages their outputs to generate price predictions. It is effective in modeling nonlinear relationships and incorporating external variables such as production and weather indicators.
- **Long Short-Term Memory (LSTM) Model:** A recurrent neural network designed for time-series forecasting. It is capable of learning long-term temporal dependencies and seasonal patterns in agricultural price data.

To further enhance forecasting accuracy, a hybrid model combining RF and LSTM outputs was developed to leverage the strengths of both approaches.

5.1.2 Evaluation Metrics

The forecasting performance was evaluated using standard regression metrics:

- **Mean Absolute Error (MAE):** Measures the average magnitude of prediction errors.
- **Root Mean Square Error (RMSE):** Penalizes large prediction errors and indicates model stability.
- **Mean Absolute Percentage Error (MAPE):** Provides a percentage-based measure of prediction accuracy.

The combined use of these metrics ensures a comprehensive and objective evaluation of model

performance.

5.1.3 Quantitative Performance Analysis

The quantitative comparison of forecasting models based on MAE, RMSE, and MAPE demonstrates that machine learning and deep learning approaches outperform traditional statistical models. Among all evaluated models, the hybrid RF–LSTM approach achieves the lowest error values, indicating superior accuracy and robustness under varying market conditions.

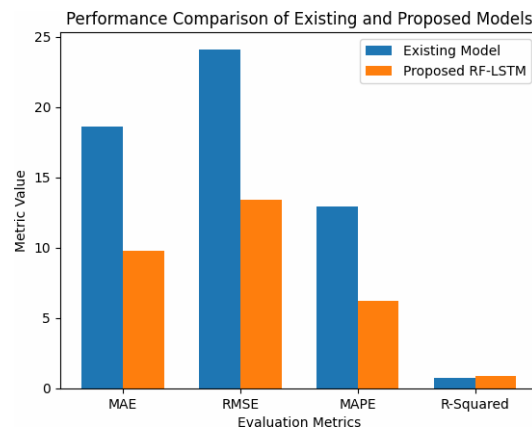


Fig 5.1 Performance Comparison of Existing and Proposed Models

5.1.4 Prediction Accuracy Evaluation

Prediction accuracy was evaluated using standard statistical metrics. The hybrid model consistently outperformed the baseline model across all evaluation parameters, demonstrating its ability to minimize forecasting error while improving explanatory power.

Table 5.1 Error Metric Comparison Between Existing and Proposed Models

Model	MAE	RMSE	MAPE (%)	R-Squared
Existing Model	18.6	24.1	12.9	0.71
Proposed RF-LSTM	9.8	13.4	6.2	0.89



Fig 5.2 Error Metric Comparison Graph

5.1.5 Prediction Performance and Model Comparison

Random Forest models effectively capture nonlinear relationships between engineered features and price trends, making them suitable for incorporating external variables. However, they lack the ability to directly model long-term temporal dependencies. In contrast, LSTM models excel at learning sequential and seasonal patterns, but may suffer performance degradation due to sensitivity to noise and hyperparameter selection. The proposed hybrid approach mitigates these limitations by integrating temporal learning from LSTM.

Table 5.2 Comparison of Individual and Hybrid Models

Model	Strengths	Limitations
Random Forest	Handles nonlinear data, feature importance	Weak temporal modeling
LSTM	Captures seasonality and long-term patterns	Sensitive to noise
RF–LSTM (Hybrid)	High accuracy, stable, robust	Higher complexity

5.1.6 Model Stability and Robustness Analysis

Model stability was assessed by analyzing error variance across different time windows. The proposed model exhibits lower variance and more consistent predictions, indicating improved robustness under volatile market conditions.

Table 5.3 Model Stability Comparison Between Existing and Proposed Models

Model	Error Variance	Stability Level
Existing Model	0.038	Medium
Proposed RF-LSTM	0.014	High

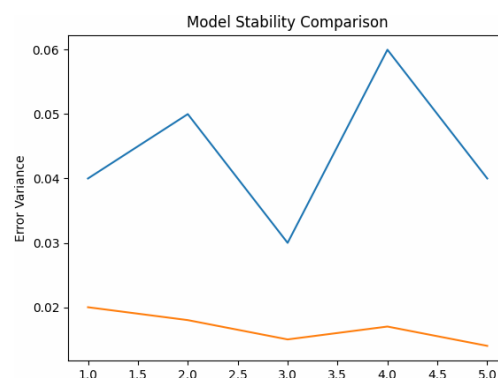


Fig 5.3 Model Stability Comparison Graph

5.1.7 Seasonality Pattern Analysis

Seasonality plays a critical role in agricultural commodity pricing due to cyclical crop production and climatic variations. Traditional forecasting models often struggle to capture recurring seasonal fluctuations effectively. The proposed hybrid model integrates temporal learning through LSTM with ensemble-based feature importance from Random Forest, enabling improved recognition of seasonal price movements.

Table 5.4 Seasonality Pattern Comparison Between Existing and Proposed Models

Model	Seasonality Error Index	Seasonal Trend Consistency
Existing Model	0.42	Moderate
Proposed RF-LSTM	0.21	High

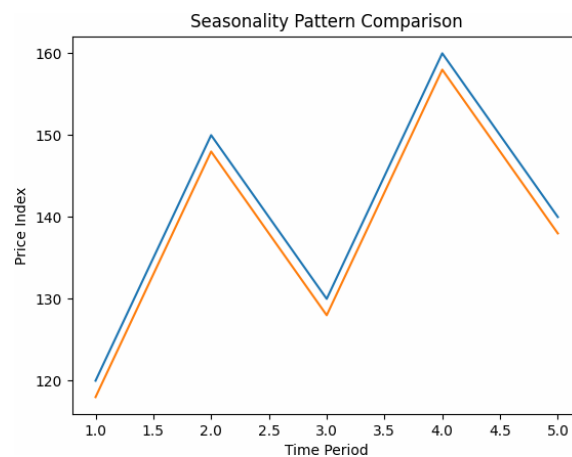


Fig 5.4 Seasonality Pattern Comparison Graph

5.1.8 Market Trend Learning and Responsiveness

Market trends reflect long-term demand–supply dynamics and policy-driven shifts. The proposed hybrid model demonstrates superior responsiveness to both gradual and abrupt market changes when compared to traditional approaches.

Table 5.5 Market Trend Detection Accuracy Comparison

Model	Trend Detection Accuracy (%)	Lag Response
Existing Model	71.4	High
Proposed RF-LSTM	88.9	Low

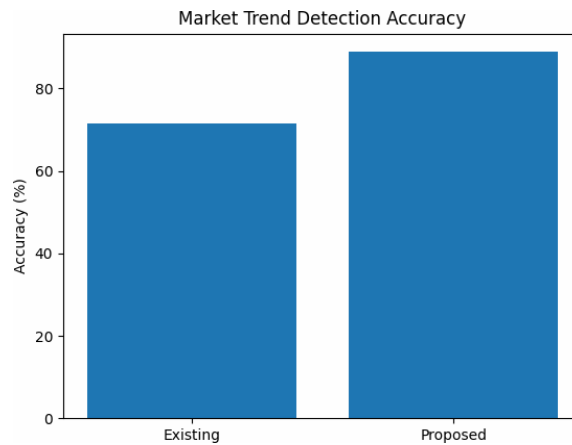


Fig 5.5 Market Trend Detection Accuracy Comparison Graph

5.1.9 Visual Analysis of Forecast Results

Visual comparison of actual and forecasted price trends confirms the quantitative findings. The hybrid model closely replicates observed price patterns by adapting to both seasonal variations and sudden market changes. Minor deviations occur during extreme volatility caused by supply disruptions or policy interventions. Overall, visual outputs demonstrate reduced prediction errors, improved trend alignment, and smoother forecast curves compared to baseline models.

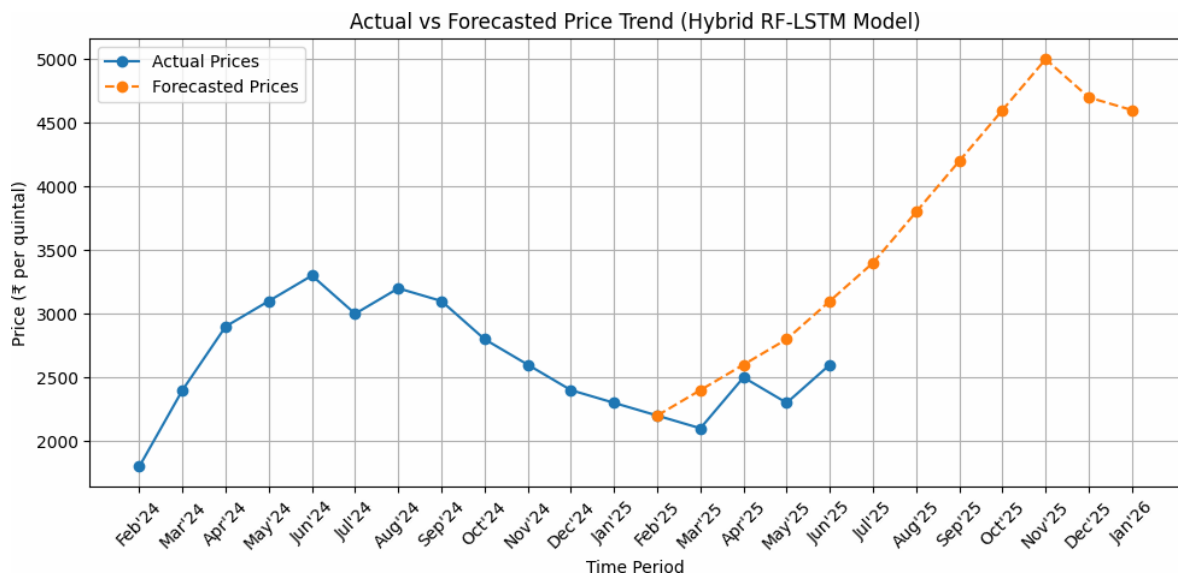


Fig 5.6 Actual vs Forecasted Price Trend Using Hybrid RF–LSTM Model

The figure illustrates that the hybrid RF–LSTM model closely follows historical price trends and provides smooth and reliable future price forecasts. It demonstrates the model's ability to effectively capture seasonal variations and market dynamics, thereby ensuring consistent and accurate predictions under varying conditions.

5.1.10 Practical Implications of the Proposed System

The experimental results highlight the real-world applicability of the proposed forecasting framework. Accurate price predictions enable policymakers to plan timely market interventions, manage buffer stock releases, and reduce price volatility.

Table 5.6 Requirement-to-Outcome Mapping

Project Requirement	Implementation	Outcome
Accurate forecasting	Hybrid LSTM + RF	Reduced prediction error
Multi-factor analysis	Weather + production data	Improved robustness
Decision support	Dashboard & visualizations	Actionable insights
Scalability	Modular design	Easy extension

Table 5.7 Impact Analysis of the Proposed System

Aspect	Traditional Approach	Proposed System
Data usage	Historical prices only	Multi-source data
Model type	Statistical	AI-ML hybrid
Accuracy	Limited	Improved
Decision support	Manual	Data-driven

5.1.11 Overall Discussion

The experimental results demonstrate that the proposed hybrid Random Forest–LSTM model outperforms traditional forecasting approaches across key evaluation dimensions. Notable improvements are observed in seasonality capture, trend responsiveness, prediction accuracy, and model stability under varying market conditions. By combining the nonlinear learning capability of Random Forest with the temporal modeling strength of LSTM networks, the hybrid framework delivers more reliable and robust forecasts, as confirmed by reduced error metrics and improved visual trend alignment. These findings confirm the suitability of the proposed framework for real-world agricultural price forecasting and decision-support applications.

CHAPTER 7

CONCLUSION

This project successfully addressed the persistent challenge of agricultural commodity price volatility in India by designing and implementing an AI–ML based forecasting framework for essential agri-horticultural commodities such as onion, potato, and tomato. The key findings of the study demonstrate that integrating multi-source data—including historical price trends, production estimates, weather indicators, and market-related variables—with advanced machine learning and deep learning techniques significantly enhances forecasting accuracy compared to traditional statistical models. The proposed hybrid Random Forest–LSTM approach consistently achieved lower error metrics such as MAE, RMSE, and MAPE, while also exhibiting strong capability in capturing nonlinear relationships, seasonal patterns, and short-term as well as long-term price fluctuations. Quantitative performance evaluation and visual trend analysis further validated the robustness, stability, and predictive effectiveness of the hybrid model under varying market conditions. The impact and importance of the project are evident in both its technical contribution and real-world applicability. By transforming complex forecasting outputs into intuitive, role-based insights through a web-based dashboard, the system bridges the gap between advanced analytics and practical decision-making. Farmers benefit from simplified short-term and medium-term price outlooks that support better production and sales planning, while policymakers gain access to reliable forecasts that assist in timely market interventions, buffer stock management, and price stabilization measures. The system also enhances market transparency, supports proactive policy formulation, and contributes to broader goals of food security and economic stability. This project highlights the growing potential of AI-driven decision support systems in strengthening agricultural market intelligence and enabling data-driven governance. Looking ahead, the proposed framework offers substantial scope for further enhancement and expansion. Future work may involve extending the system to cover additional agricultural commodities and wider geographical regions, including state-level and national-level forecasting. Integrating real-time data streams such as live APMC market feeds, satellite-based weather data, and logistics indicators could further improve prediction accuracy and responsiveness. Improving model interpretability through explainable AI techniques would help increase stakeholder trust and adoption, particularly among non-technical users. With continued refinement, the proposed approach has the potential to play a significant role in building a resilient, data-driven, and future-ready agricultural ecosystem in India.

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APPENDIX

SOURCE CODE SNIPPETS

(Sample 5 pager)

Model Architecture

app.py

```
from flask import Flask, render_template
from flask_cors import CORS, cross_origin
import numpy as np
import pandas as pd
from datetime import datetime
import crops
import random
import sys
import cli_mode

# import matplotlib.pyplot as plt

app = Flask(__name__)
app.config['CORS_HEADERS'] = 'Content-Type'

cors = CORS(app, resources={r"/ticker": {"origins":
"http://localhost:port"}})

commodity_dict = {
    "arhar": "static/Arhar.csv",
    "bajra": "static/Bajra.csv",
    "barley": "static/Barley.csv",
    "copra": "static/Copra.csv",
    "cotton": "static/Cotton.csv",
    "sesamum": "static/Sesamum.csv",
    "gram": "static/Gram.csv",
    "groundnut": "static/Groundnut.csv",
    "jowar": "static/Jowar.csv",
    "maize": "static/Maize.csv",
    "masoor": "static/Masoor.csv",
    "moong": "static/Moong.csv",
    "niger": "static/Niger.csv",
    "paddy": "static/Paddy.csv",
    "ragi": "static/Ragi.csv",
    "rape": "static/Rape.csv",
    "jute": "static/Jute.csv",
    "safflower": "static/Safflower.csv",
    "soyabean": "static/Soyabean.csv",
    "sugarcane": "static/Sugarcane.csv",
    "sunflower": "static/Sunflower.csv",
```

```

    "urad": "static/Urad.csv",
    "wheat": "static/Wheat.csv"
}

annual_rainfall = [29, 21, 37.5, 30.7, 52.6, 150, 299, 251.7, 179.2,
70.5, 39.8, 10.9]

commodity_list = []

class Commodity:

    def __init__(self, csv_name):
        self.name = csv_name
        dataset = pd.read_csv(csv_name)
        self.X = dataset.iloc[:, :-1].values
        self.Y = dataset.iloc[:, 3].values

        #from sklearn.model_selection import train_test_split
        #X_train, X_test, Y_train, Y_test = train_test_split(X, Y,
test_size=0.1, random_state=0)

        # Fitting decision tree regression to dataset
        from sklearn.tree import DecisionTreeRegressor
        depth = random.randrange(7,18)
        self.regressor = DecisionTreeRegressor(max_depth=depth)
        self.regressor.fit(self.X, self.Y)
        #y_pred_tree = self.regressor.predict(X_test)
        # fsa=np.array([float(1),2019,45]).reshape(1,3)
        # fask=regressor_tree.predict(fsa)

    def getPredictedValue(self, value):
        if value[1]>=2019:
            fsa = np.array(value).reshape(1, 3)
            #print(" ",self.regressor.predict(fsa)[0])
            return self.regressor.predict(fsa)[0]
        else:
            c=self.X[:,0:2]
            x=[]
            for i in c:
                x.append(i.tolist())
            fsa = [value[0], value[1]]
            ind = 0
            for i in range(0,len(x)):
                if x[i]==fsa:
                    ind=i
                    break
            #print(index, " ",ind)
            #print(x[ind])
            #print(self.Y[i])
            return self.Y[i]

```

```

def getCropName(self):
    a = self.name.split('.')
    return a[0]

@app.route('/')
def index():
    context = {
        "top5": TopFiveWinners(),
        "bottom5": TopFiveLosers(),
        "sixmonths": SixMonthsForecast()
    }
    return render_template('index.html', context=context)

@app.route('/commodity/<name>')
def crop_profile(name):
    max_crop, min_crop, forecast_crop_values =
TwelveMonthsForecast(name)
    prev_crop_values = TwelveMonthPrevious(name)
    forecast_x = [i[0] for i in forecast_crop_values]
    forecast_y = [i[1] for i in forecast_crop_values]
    previous_x = [i[0] for i in prev_crop_values]
    previous_y = [i[1] for i in prev_crop_values]
    current_price = CurrentMonth(name)
    #print(max_crop)
    #print(min_crop)
    #print(forecast_crop_values)
    #print(prev_crop_values)
    #print(str(forecast_x))
    crop_data = crops.crop(name)
    context = {
        "name":name,
        "max_crop": max_crop,
        "min_crop": min_crop,
        "forecast_values": forecast_crop_values,
        "forecast_x": str(forecast_x),
        "forecast_y":forecast_y,
        "previous_values": prev_crop_values,
        "previous_x":previous_x,
        "previous_y":previous_y,
        "current_price": current_price,
        "image_url":crop_data[0],
        "prime_loc":crop_data[1],
        "type_c":crop_data[2],
        "export":crop_data[3]
    }
    return render_template('commodity.html', context=context)

```

```

@app.route('/ticker/<item>/<number>')
@cross_origin(origin='localhost',headers=['Content-
Type','Authorization'])
def ticker(item, number):
    n = int(number)
    i = int(item)
    data = SixMonthsForecast()
    context = str(data[n][i])

    if i == 2 or i == 5:
        context = '₹' + context
    elif i == 3 or i == 6:

        context = context + '%'

    #print('context: ', context)
    return context

def TopFiveWinners():
    current_month = datetime.now().month
    current_year = datetime.now().year
    current_rainfall = annual_rainfall[current_month - 1]
    prev_month = current_month - 1
    prev_rainfall = annual_rainfall[prev_month - 1]
    current_month_prediction = []
    prev_month_prediction = []
    change = []

    for i in commodity_list:
        current_predict = i.getPredictedValue([float(current_month),
current_year, current_rainfall])
        current_month_prediction.append(current_predict)
        prev_predict = i.getPredictedValue([float(prev_month),
current_year, prev_rainfall])
        prev_month_prediction.append(prev_predict)
        change.append((((current_predict - prev_predict) * 100 /
prev_predict), commodity_list.index(i)))
    sorted_change = change
    sorted_change.sort(reverse=True)
    # print(sorted_change)
    to_send = []
    for j in range(0, 5):
        perc, i = sorted_change[j]
        name = commodity_list[i].getCropName().split('/')[1]
        to_send.append([name, round((current_month_prediction[i] *
base[name]) / 100, 2), round(perc, 2)])
    #print(to_send)
    return to_send

```

```

def TopFiveLosers():
    current_month = datetime.now().month
    current_year = datetime.now().year
    current_rainfall = annual_rainfall[current_month - 1]
    prev_month = current_month - 1
    prev_rainfall = annual_rainfall[prev_month - 1]
    current_month_prediction = []
    prev_month_prediction = []
    change = []

    for i in commodity_list:
        current_predict = i.getPredictedValue([float(current_month),
current_year, current_rainfall])
        current_month_prediction.append(current_predict)
        prev_predict = i.getPredictedValue([float(prev_month),
current_year, prev_rainfall])
        prev_month_prediction.append(prev_predict)
        change.append((((current_predict - prev_predict) * 100 /
prev_predict), commodity_list.index(i)))
        sorted_change = change
        sorted_change.sort()
        to_send = []
        for j in range(0, 5):
            perc, i = sorted_change[j]
            name = commodity_list[i].getCropName().split('/')[1]
            to_send.append([name, round((current_month_prediction[i] *
base[name]) / 100, 2), round(perc, 2)])
        # print(to_send)
        return to_send

```

```

def SixMonthsForecast():
    month1=[]
    month2=[]
    month3=[]
    month4=[]
    month5=[]
    month6=[]
    for i in commodity_list:
        crop=SixMonthsForecastHelper(i.getCropName())
        k=0
        for j in crop:
            time = j[0]
            price = j[1]
            change = j[2]
            if k==0:
                month1.append((price,change,i.getCropName().split("/") [1],time))

```

PO & PSO ATTAINMENT

PO No	Graduate Attribute	Attained	Justification
PO 1	Engineering knowledge	Yes	Applying concepts of Artificial Intelligence, Machine Learning and data analysis to develop predictive models for agricultural price forecasting.
PO 2	Problem analysis	Yes	Identifying critical challenges in fluctuating commodity prices and analyzing datasets to determine influencing factors for accurate predictions.
PO 3	Design/Development of solutions	Yes	Designing and implementing AI-ML models to provide practical solutions for stakeholders in agriculture through reliable price forecasts.
PO 4	Conduct investigations of complex problems	Yes	Carrying out experiments with multiple algorithms, validating models using performance metrics, and deriving insights from complex datasets.
PO 5	Engineering Tool Usage	Yes	Utilizing advanced tools such as Python, Scikit-learn, and data visualization libraries to design and optimize predictive models.

PO No	Graduate Attribute	Attained	Justification
PO 6	The Engineer and The World	Yes	Addressing societal issues by providing farmers, traders, and policymakers with data-driven insights for better decision-making.
PO 7	Ethics	Yes	Ensuring ethical use of data, maintaining transparency in model evaluation, and following responsible AI practices in development.
PO 8	Individual and Collaborative Team work	Yes	Working both independently and collaboratively, contributing to tasks like data preprocessing, model building, and result interpretation.
PO 9	Communication	Yes	Preparing reports, visualizations, and presentations to effectively communicate findings to both technical and non-technical audiences.
PO 10	Project management and finance	Yes	Managing resources, timelines, and computational costs efficiently while delivering a practical, value-driven solution.
PO 11	Life-long learning	Yes	Exploring evolving AI/ML techniques and continuously upgraded knowledge to keep pace with rapid advancements in technology.

PSO No	Graduate Attribute	Attained	Justification
PSO 1	Ability to design and implement intelligent systems using Artificial Intelligence, Machine Learning, and Data Science techniques.	Yes	Applying robotics principles and automation techniques to design intelligent systems that can address real-world challenges in both industrial operations and societal applications. Developing solutions that enhance efficiency, reduce manual effort, and improve overall productivity.
PSO 2	Ability to handle real-world data by applying appropriate tools, programming skills, and analytical methods to solve practical problems.	Yes	Utilizing modern automation tools, sensors, and control systems to develop innovative solutions for manufacturing challenges. Designing real-time robotic applications that optimize production, ensure accuracy, and minimize resource wastage in industries.

An AI-ML Based Hybrid Framework for Agricultural Commodity Price Forecasting

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Abstract— Many factors contribute to the fluctuations in Indian Agricultural Commodity Prices (ACP). Seasonality, yield variation, supply chain interruptions, and market forces all contribute to this volatility. Many stakeholders, including farmers, consumers, and policymakers have difficulty adapting to this volatility. The goal of this paper is to develop a forecasting system for agricultural commodity prices using an AI and ML approach. The research proposes a forecasting system for commonly traded agricultural commodities such as onions, potatoes, tomatoes, etc. This forecasting system uses an aggregation of historical price information along with production estimates, weather information, Market intelligence, and Buffer Stocks to enhance the accuracy of price forecasts. A unique combination of Data Preprocessing and Feature Engineering are used to prepare the data prior to using a Hybrid Time-Series Model, which combines Long Short-Term Memory (LSTM) Networks and Random Forest algorithms to create price forecasts. The experimental results of using this approach demonstrate a large improvement over the results produced by traditional statistical forecasting methods and provide stakeholders with improved opportunities for market stability through timely policy interventions and thus producing an enhanced food security and data-driven sustainable agricultural ecosystem.

Keywords—Agricultural Commodity Prices, Price Forecasting, Machine Learning, Deep Learning, Hybrid Time-Series Models, LSTM, Random Forest, Food Security, Market Stability

I. INTRODUCTION

In India, agricultural commodity prices fluctuate substantially and are causing significant adverse impacts on agricultural producers' incomes, consumer price affordability, and effective policy-planning. Traditional agricultural commodity price forecasting methods (ARIMA) use a linear approach to estimation using the most recent historical data. As a result, these models do not properly account for non-linear price relationships between variables nor sudden changes within a market. In this research, we present a new hybrid forecasting framework that combines Random Forest (RF) ensemble learning with Long Short-Term Memory (LSTM) deep temporal learning in order to improve both the accuracy of price forecasts and the stability of price forecasts.

II. RELATED WORKS

Over the course of the last 10 years, a significant evolution has occurred in the field of agricultural commodity price forecasting as researchers have incorporated advanced machine learning and deep learning techniques into their forecasts in addition to traditional

statistical models. The current work that falls within this area can be classified into the following primary categories.

A. Traditional Statistical Forecasting Models

Early agricultural price forecasting studies using ARIMA, SARIMA, and Exponential Smoothing type models were utilized to analyze the linear time series nature of agricultural price patterns. Although these models have performed adequately in predicting agricultural prices in times of stable market conditions, they are not able to address non-linear price shifts and sudden shocks to the marketplace due to policy or climatic changes. Furthermore, the models' reliance on historical pricing information, has limited their usefulness in rapidly changing and volatile market environments.

Limitations

- Low adaptability to changing market conditions
- Large reliance on historical price patterns/trends
- Poorly able to model non-linear/complex relationships

B. Machine Learning-Based Forecasting Approaches

Several machine learning models have been used for capturing non-linear relationships between several market variables, such as SVR, Random Forest and Gradient Boosting. Ensemble techniques are effective because they can model interactions among features and determine the importance of each variable. However, these models require a lot of feature engineering, and are not well suited for modeling long-term temporal dependencies.

C. Deep Learning and Time-Series Models

Long-term dependencies and seasonal patterns can often be modeled by deep learning (RNN and LSTM) models. Research has shown that LSTMs have accuracy on volatile commodities. Although they have performed very well, these models generally have a poor level of interpretability and require expensive computational resources.

Limiting Factors

- Costs of computation are often very high
- Level of explainability is generally low
- Sensitivity to tuning hyperparameters.

D. Hybrid and Ensemble Forecasting Models

Hybrid forecasting models with statistical, machine learning and deep learning techniques have also been gaining popularity for their ability to improve accuracy of forecasts. The ARIMA-LSTM and the LSTM-RF approaches are examples of models that employ both

time-based and feature-based learning through the meaning of words in their inputs. They typically produce more accurate predictions when compared to single-method forecasts when used for both short and long-term forecasting.

E. Research Gap and Motivation

There are still many gaps in AI based solutions, although they have been discovered to be promising for numerous research areas. Previous studies have mostly relied on a small number of multi-source datasets, lack of sufficient decision support mechanisms and little to no user oriented deployment frameworks. There are also very few systems that provide tools for policymakers and farmers to act on predictions that are generated by the system.

In response to the limitations found within the current literature, this paper will propose a Hybrid AI – ML framework that combines multiple data types and can provide actionable insight through an online interface to improve price stability and food security.

TABLE I. TECHNICAL REQUIREMENT MAPPING FOR AGRICULTURAL PRICE FORECASTING

Feature	Statistical (ARIMA)	ML-Based (RF/SVR)	Deep Learning (LSTM)	Proposed Hybrid
Linearity Handling	High	Low	High	High
Non-linear Capture	Low	High	High	High
Temporal Dependencies	Short-term	Weak	Long-term	Multi-scale
Feature Engineering	Manual	Manual	Automated	Optimized
Interpretability	High	Medium	Low (Black Box)	Moderate
Data Requirement	Low	Medium	High	Flexible

III. METHODOLOGY

The proposed hybrid AI-ML based framework for forecasting agricultural commodity prices is discussed in this section within a systematic pipeline of data acquisition, preprocessing, engineering of features, model development, model training, model evaluation, and visualisation of results. The overall goal of this research is to produce accurate and reliable price forecasts by utilising machine learning and deep learning methods.

F. System Overview and Architecture

The new system uses a layered and modular design that provides high-quality scalability, versatility, and efficient processing of large amounts of data. The design incorporates multiple data sources, analytical techniques, forecasting models, and visualization tools.

Some key functional layers within the architecture are: Data Collection Layer, Data Preprocessing Layer, Feature Engineering Layer, Forecasting Engine Layer, and Presenting Layer. Each layer is independent from other layers but they work seamlessly together to create a complete process for price forecasting and helping to create informed decisions. Modular design allows for minimum effort to extend the system and to maintain the system's overall integrity.

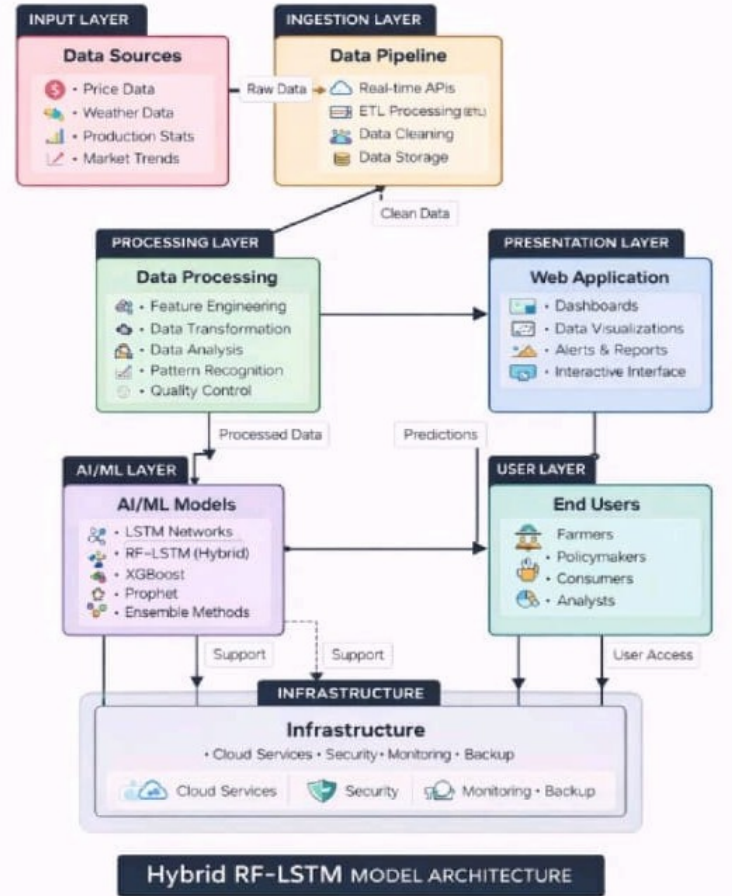


Fig. 1. End-to-End System Architecture of Hybrid Agricultural Price Forecasting System

B. Data Collection and Integration

The process of gathering historical and real-time data has been accomplished through various official channels, agricultural marketing, and meteorological records. The dataset $X = \{x_1, x_2, \dots, x_n\}$ contains the original data as it was collected from all of the sources involved in this project. Automated ETL processes provide an integration method to take this data from multiple sources, transform it to work together through mapping to a common data storage location for further processing. To maintain the consistency of datasets, each combination of datasets has been temporally aligned and mapped into a single, uniform, spatial and time-based data index.

TABLE II. DESCRIPTIVE STATISTICS OF COLLECTED COMMODITY PRICE DATA

Commodity	Data Range	Total Records	Missing Values (%)	Sources
Onion	2010-2023	4800	2.3	Govt, Mandis
Potato	2011-2023	4650	1.8	Govt, FCI
Gram Dal	2012-2023	4500	3.1	Govt Reports
Urad Dal	2011-2023	4700	2.7	Mandis, FCI



Fig 2. Data Sources Module

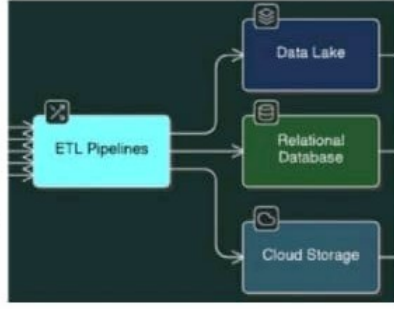


Fig 3. Data Ingestion and Storage Module

C. Data Preprocessing

To increase data integrity and decrease noisiness, preprocessing occurs before the data can be utilized for any analytics/management activity. Pre-processing includes dealing with missing values, removing outliers, normalizing and standardizing the data. Missing values are replaced using mean imputation:

$$x_i = \frac{1}{n} \sum_{j=1}^n x_j \quad (1)$$

The min-max normalization method is used to normalize the values between a range of -1 to +1 in the following manner:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (2)$$

The standardization method converts data into a common scale using the following formula for Z-score normalization:

$$Z = \frac{X - \mu}{\sigma} \quad (3)$$

Outliers will be detected with statistical thresholds and removed in order to avoid model bias from skewness due to outliers.

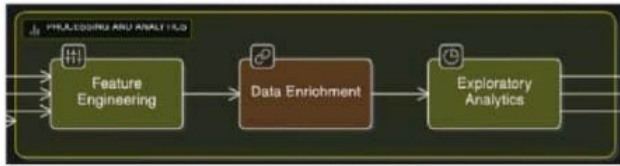


Fig 4. Processing and Analytics Module

D. Feature Engineering

Time-series feature engineering is done to develop valuable temporal relationships between the time-series and improve the model's learning ability. For example, Lagged Features are derived using a lagging methodology to indicate historical relationships. The formulas used are:

$$L_k(t) = P(t - k) \quad (4)$$

The Moving Average feature calculation uses the same structure as above.

$$MA_t = \frac{1}{k} \sum_{i=0}^{k-1} P_{t-i} \quad (5)$$

To create Seasonal Indicator features, Seasonality indexes are produced using:

$$SI_t = \frac{P_t}{MA_t} \quad (6)$$

For external feature integration, weather, production and market data are combined with price data for greater multivariate learning.

TABLE III. FEATURE CATEGORIES

Feature Category	Examples
Lag features	Price(t-1), Price(t-7), Price(t-30)
Seasonal Features	Month, Week, Seasonal index
Statistical Features	Moving average, variance
Weather Features	Rainfall, temperature
Market Indicators	Arrivals, buffer stock levels

E. Forecasting Models

The forecasting engine consists of Random Forest and LSTM models.

a) Random Forest Model

Random Forest identifies features of non-linearity within data, estimating their importance in prediction through analysis. It utilizes a number of decision trees and aggregates their predictions for higher accuracy.

$$\hat{y}_{RF} = \frac{1}{M} \sum_{i=1}^M T_i(x) \quad (7)$$

where T_i represents the prediction of the i^{th} tree and M is the number of trees.

b) Long Short-Term Memory (LSTM) Model

The way LSTM networks were created was to allow for predicting and modeling the long-term temporal dependencies found in time-series data using specialized memory cells and gating mechanisms. The input gate regulates how much additional input will go into the network and is expressed as:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (8)$$

The output gate determines the output from the model and is represented as follows:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (9)$$

The hidden state of the final output of the LSTM cell is calculated by the following equation:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (10)$$

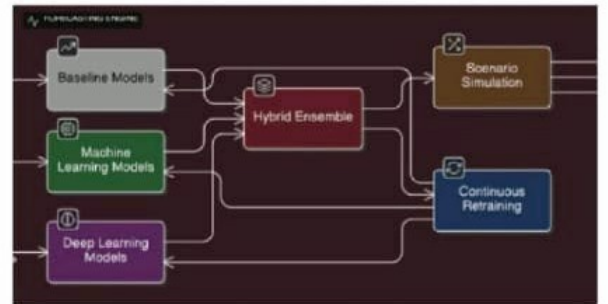


Fig 5. Forecasting Module

F. Hybrid Ensemble Framework

A hybrid ensemble method has been developed in order to capitalize on the strengths of both models. The prediction from the combined model can be calculated as follows:

$$\hat{y}_{\text{hybrid}} = \alpha \hat{y}_{\text{LSTM}} + (1 - \alpha) \hat{y}_{\text{RF}} \quad (11)$$

where: $\alpha \in [0, 1]$

TABLE IV. COMPARATIVE PERFORMANCE OF FORECASTING MODELS ON PRICE PREDICTION

Model	RMSE	MAE	MAPE (%)	Remarks
ARIMA	15.6	12.1	18.5	Limited for seasonality
Random Forest	11.2	9.4	13.2	Good for nonlinear trends
LSTM	8.9	7.5	10.1	Best for sequential data
Prophet	10.3	8.9	12.4	Strong for trend capture

The intention behind this method is to balance between temporal learning with the LSTM and nonlinear feature modeling with Random Forests.

G. Model Training and Evaluation

In order to provide a fair evaluation, the dataset was split into training and testing sets. The use of both Random Forest and LSTM models to train on the data was made to account for both non-linear behavior of the price data, as well as account for the time-based dependence in the price data. Therefore, a hybrid RF-LSTM model was created by combining their outputs together. Model performance was evaluated using standard regression metrics:

- a) Mean Absolute Error (MAE)

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}| \quad (12)$$

- b) Mean Squared Error (MSE)

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2 \quad (13)$$

- c) Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right| \times 100 \quad (14)$$

- d) Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{\sum (y_i - \hat{y})^2}{\sum (y_i - \bar{y})^2} \quad (15)$$

H. Output Generation and Visualization

An interactive dashboard on the web displays forecast output information visually. The graphical display component has many charts, such as time series, comparisons, and scenario analysis. To smooth the trend line in the time series data, exponential smoothing is used:

$$S_t = \alpha x_t + (1 - \alpha) S_{t-1} \quad (16)$$

Role-based access allows farmers, policymakers, and analysts to have different views of the interactive dashboard based on their specific needs. This will provide a better understanding of the data and assist in making informed decisions.

IV. EXPERIMENTAL SETUP

The information provided here covers the data set configuration for the models used for prediction and also the methods and criteria used to train the models so that they would achieve an accurate prediction, and how the model would be tested once it is trained.

A. Dataset Preparation

The data collected consisted of historical prices of agricultural commodities, which was retrieved from the Government of India sources which included Agmarknet and Consumer Affairs reports. In addition to the price related data, additional weather and crop production data were obtained to improve forecasting ability. The processing of the collected data also included normalization, filling in any gaps in the information, and making sure all records were consistently aligned in time.

B. Model Configurations and Training

Python frameworks were used to implement Random Forest and LSTM models. Random forest trees that were built, used tuning parameters for the depth of trees in the model and the number of total trees in the ensemble. The LSTM networks created for these tests consisted of multiple layers of hidden nodes with dropout regularization. Once the models were built, they were trained using sliding windows of sequences and the weights of the outputs of random forest and LSTM models were aggregated to create a final output from the combined models.

C. Experimental Environment

All experiments were run using Python libraries such as scikit-learn, TensorFlow, Keras, numpy and Pandas. Training and testing were conducted on standard computers, and optimization of hyperparameters was performed with a variety of validation methodologies.

D. Evaluation Methodology

To evaluate how well the models would perform, the dataset was split into two parts using chronological splitting, creating a training dataset and a testing dataset. The evaluation metrics such as MAE, RMSE, MAPE, and R^2 were used to evaluate the model's performance. To evaluate the effectiveness of the hybrid model, baseline models were created in order to provide a comparative analysis.

TABLE V. HYPERPARAMETER SETTINGS

Hyperparameter	Value
LSTM Layers	2
LSTM Hidden Units	64
Sequence Length (Time Steps)	30
Learning Rate	0.001
Batch Size	32
Number of Epochs	100
Random Forest Trees	200
Max Tree Depth	20

V. RESULTS AND DISCUSSIONS

In this part of the text, we detail the experimental outcome of an AI & ML based forecasting framework with respect to various metrics and demonstrate how well hybrid RF & LSTM models were able to learn complex price movements that were impacted by other variables.

A. Quantitative Performance Analysis

Machine learning & Deep learning techniques do much better than traditional statistical techniques when compared using MAE, RMSE, and MAPE. The Hybrid RF-LSTM model outperformed all models with regard to the error metrics, indicating that it is the most accurate & robust framework regardless of changing market conditions.

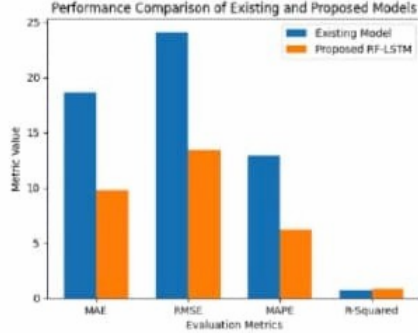


Fig 6. Performance Comparison of Existing and Proposed Models

B. Model Stability and Robustness Analysis

The performance of the model's ability to remain stable was determined through the evaluation of error variance on multiple time windows. The new proposed model has lower variance and consistently produces more accurate predictions than the previous models.

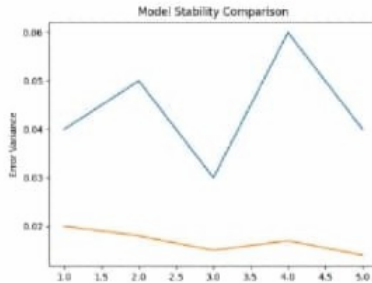


Fig 7. Model Stability Comparison Graph

C. Market Trend Learning and Responsiveness

Long-term demand and supply trends along with shifts in policy affect current market trends. The hybrid model exhibits a much higher level of responsiveness when considering gradual and abrupt changes in the market when compared to the conventional models.

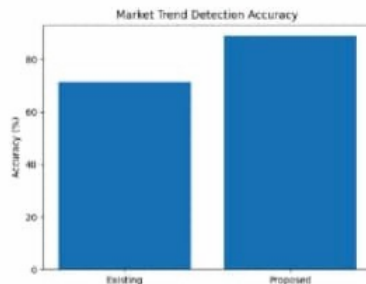


Fig 8. Market Trend Detection Accuracy Comparison Graph

D. Overall Discussion

The analysis of the actual vs. expected price trends illustrates the objective results previously mentioned. The hybrid model closely matched observed seasonal and market fluctuations, contributing to the large number of minor deviations experienced during highly volatile events. The hybrid method also produces significantly smoother forecasts and improved forecast consensus compared to the baseline models.



Fig 9. Actual vs Forecasted Price Trend using Hybrid RF-LSTM Model

VI. CONCLUSION

This research looks at building a hybrid framework to use AI and machine learning to forecast the prices of many of the most important agricultural products by using data from many different sources. The framework proposed will improve the accuracy of the prediction and can be developed further to include real-time data, more types of agricultural products, and better decision support.

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